The Folded Space

Fold, Return and Closure:
a Treatise on the FTM World Model

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Abstract

Space is described as a closed, folded three-dimensional manifold, pictured as a funnel. Mass is fold, $\Delta s = GM/(3c^2)$. Slope, clock rate and energy follow from it. Because space has no outside, every motion returns in an inner direction that closes at the closure length Λ. From this follow a threshold $a_0 = c^2/(2\Lambda)$ and a rule of motion, $a = a_N/(1 - e^{-\sqrt{(a_N/a_0)}})$, which is Newtonian above the threshold and gives flat rotation curves below it, without additional mass. The rotation curves of 141 galaxies (SPARC) read $a_0 = 1.11 \cdot 10^{-10}$ m/s^2, the scale of the early sound wave (DESI) 1.10, the shape of the distances of supernovae and BAO 1.09: the same threshold for part and whole. The breathing after Planck reads 1.065. Dark energy is the restoring of the three loops of the torus. From a_0 and the measured densities of the baryons and the radiation follow $H_0 = 67.2$ km/s/Mpc and $\Omega_{DE} = 0.685$. All levels are ordered by three symmetries of the torus shape, mirror, threefold and full turn, and by two ladders. Nucleus, atom, substances, stars and galaxies show the same forms, tested on nuclear masses (AME2020), on the periodic table (PubChem) and on the galaxies.

Preface: Terms Are Tools

This treatise describes the world with a few images: funnel, fold, slope, clock rate, return, loop, closure. These words are tools, not new things. Behind each of them stands a mathematical object that geometry named long ago. The most important one is the **manifold**: a space that looks like ordinary flat space in the small, but may be curved, folded and closed in on itself in the large. The surface of the Earth is a two-dimensional manifold: every piece looks flat like a sheet of paper, while the whole is a sphere without an edge.

Fold and **funnel** are, in this model, the same as a curved three-dimensional manifold, only named more vividly. We keep our terms because they help to explain: a funnel can be drawn, a metric cannot. Wherever a term has its own precise meaning in the model, it is explained where it first appears. All symbols are collected in Appendix A.

One rule holds throughout: nothing is explained as an additional substance. Everything observed is a state or a rule of the one medium, space itself.

The course of the treatise. Part I starts with the funnel: what the fold is, and how slope, clock rate and energy follow from it. Part II closes space: because it has no outside, every motion returns, and the threshold and the rule follow from that return. Part III shows the symmetries that carry the whole, the mirror, the threefold and the full turn, and the two ladders on which all levels are ordered. Part IV goes through the levels from the particle to the galaxy, Part V through the whole with threshold, breathing and dark energy. Part VI explains how to read measurements without mistaking rules for things, what can be tested and what remains open. Every step builds only on the previous ones.

Part I: The Funnel

1 The Funnel as an Image of the Folded Manifold

We live in three spatial directions. We cannot look at a curvature of this three-dimensional space from outside, because there is no outside from which to look. One helps oneself with a cut: take a flat plane through the middle of a body, say the Sun, and draw how long the paths in this plane really are. Where space is folded, the paths inward are longer than flat geometry would lead us to expect. Drawing this extra length as depth pointing downward produces a **funnel**.

The funnel therefore shows the three-dimensional manifold in two dimensions. Its depth is not a place one could fall into, but a measure: a funnel that sinks deeper represents **more space within space**. Between two circles of equal circumference, the funnel holds a longer path than the flat plane. Space contains more there than its boundary reveals.

2 The Fold Δs

The **fold** Δs is the extra length in the centre of a body that folded space contains compared with flat space. It is the measure of mass:

$$\Delta s = GM/(3c^2)$$

Derivation. The fold grows with the mass, and it is distributed over the body: for a body of mass M, volume V, surface S and radius R, the extra radial length is $\Delta s = (GM/c^2)\cdot V/(S\cdot R)$, the length GM/c^2 weighted by the ratio of content to boundary. In the language of general relativity this is the excess radius of a homogeneous sphere. The ratio $V/(S\cdot R)$ is a pure shape number: for the sphere, $V = (4/3)\pi R^3$ and $S = 4\pi R^2$, so $V/(S\cdot R) = 1/3$. Hence the 3 in the denominator: it is the spatial dimension $d = 3$. A homogeneous sphere in d dimensions has $\Delta s = GM/(d\cdot c^2)$.

Examples. The Earth carries a fold of $\Delta s = 1.478$ mm, the Sun of $\Delta s = 492.2$ m, the Milky Way with $6\cdot 10^{10}$ solar masses of $\Delta s = 197.4$ AU. The fold depends only on the mass, not on the radius: a star that inflates keeps its fold.

3 The Slope g

The **slope** is the steepness of the funnel wall at a place, what we feel as weight:

$$g = GM/r^2 = 3c^2\cdot\Delta s/r^2$$

It follows from the fold by inserting $GM = 3c^2\Delta s$. The product $g\cdot r^2 = 3c^2\Delta s$ is fixed: the fold sits in the centre, and the slope falls off outward with the square of the distance. If a body expands at fixed mass, its edge flattens while the fold in the centre remains (the balloon effect): a red giant is deeply folded and only weakly sloped at its surface.

4 The Clock Rate and Time

The **clock rate** $d\tau/dt$ is the ratio in which space unrolls at a place, compared with flat outer space. It follows from the fold, since the clock rate at distance r is $\sqrt{(1 - 2GM/(rc^2))}$, and $2GM/c^2 = 6\Delta s$:

$$d\tau/dt = \sqrt{(1 - 6\Delta s/r)} \qquad \delta = 1 - d\tau/dt \approx 3\Delta s/r$$

Time is the difference in unrolling between folded and flat regions, not friction and not a flow of its own. It arises only where space is folded. Between two places it is $\delta_{AB} = d\tau/dt|_B - d\tau/dt|_A$. The small quantity δ is called the **spread**: the distance of the clock rate from its flat value of one.

At the **limit radius** $r_0 = 6\Delta s$ the clock stands still. For the Earth it lies at 8.87 mm, for the Sun at 2.95 km. A clock on the Earth's surface falls behind flat outer space by 0.022 s per year, on the Sun's surface by 67 s. In the Sun's funnel at the Earth's orbit the figure is 0.31 s. On a neutron star (1.4 M☉, 12 km) the clock rate is 0.81.

Light in the clock rate. c is the scale of the unfolded envelope: $c^2 = GM/(3\Delta s)$ links fold and slope. Measured at the place, light is always c, because clock and light follow the same rate. Counted from outside it is slower, $c_{eff} = c\cdot\sqrt{(1 - 6\Delta s/r)}$, a delay of the unrolling, not friction.

5 Energy Is Bound Fold

Mass and energy are the same, and both are fold:

$$E = Mc^2 = 3c^4\cdot\Delta s/G$$

Every energy, whether rest, motion or field, is bound fold. When a body binds to another, it gives up fold.

Derivation of binding. On a circular orbit in the funnel, $v^2 = GM/r = 3c^2\Delta s/r$, hence $\delta = 3\Delta s/r = v^2/c^2$: the spread of an orbit is the square of its speed relative to c. The binding energy of a circular orbit is $\frac{1}{2}mv^2$, that is $\frac{1}{2}\cdot\delta$ of the rest energy. **Binding gives up half the spread.**

Examples. In the hydrogen atom the electron circulates with $v/c = \alpha$, the fine-structure constant. The binding is therefore $\alpha^2/2$ of the electron's rest energy, $2.661\cdot10^{-5}$ including the co-motion of the nucleus. The measured 13.598 eV (PubChem) is exactly this fraction. The iron nucleus ^{56}Fe carries 0.94 % less mass than its nucleons would carry separately (AME2020).

6 Orbits and Light in the Funnel

A planet follows the slope that the Sun's fold produces at its orbital radius:

$$v = c\cdot\sqrt{(3\Delta s_\odot/r)} \qquad T^2 = 4\pi^2\cdot r^3/(3c^2\cdot\Delta s_\odot)$$

With $\Delta s_\odot$ the fold of the Sun this gives for the Earth the orbital speed v = 29.78 km/s and the period T of one year. Every motion in the funnel feels two things: the slower clock rate and the extra length. For light both parts are equal:

$$\text{deflection} = 6\Delta s/b \text{ (clock rate)} + 6\Delta s/b \text{ (fold)} = 12\Delta s/b \qquad \text{solar limb: } 0.8756'' + 0.8756'' = 1.7512''$$

Here b is the distance at which the ray passes the centre. Orbits feel both too: the perihelion of an orbit advances by $(12\pi + 12\pi - 6\pi)\cdot\Delta s/p = 18\pi\Delta s/p$ per revolution, with p the parameter of the orbit, its half width through the focus. For Mercury that is 42.98″ per century. Expressed with the spread $\delta = 3\Delta s/p$: the orbit misses the full turn by 3δ, and light is deflected by 4δ.

The funnel has fixed **steps**: at $r_0 = 6\Delta s$ the clock stands still, at $9\Delta s$ lies the photon sphere, at $18\Delta s$ the innermost stable orbit (Sun 4.43 and 8.86 km). The circles $6\Delta s$, $9\Delta s$, $18\Delta s$ are 2d, 3d, 6d with $d = 3$: the whole numbers of the funnel are multiples of the dimension.

Part II: The Closed Space

7 No Outside: the Torus

Space has no edge and no outside. It loops back into itself. The simplest form of such a space in three directions is the **torus T^3**: take a cube and declare opposite faces to be the same. Whoever walks out on the right comes back in on the left. Every direction therefore has a finite **loop**, a closed path that leads around space. The torus T^3 has three independent loops.

Because there is no outside, fold cannot drain away anywhere. When funnels fold extra length into their centres, the closed medium answers with a stretching of the loops:

$$\sigma = \Sigma\Delta s_{\text{centres}} - \Delta L_{\text{space}} = 0$$

The medium thus runs a cycle: flat matrix ($\Delta s = 0$), vortex (slope g), fold (extra length Δs in the centre), return through the closed whole. Charges, by contrast, cancel on the closed space, $\Sigma Q_{el} = 0$. And the form of the slope is constrained: the torus has Euler characteristic zero, so every slope obeys funnels − filament saddles + wall saddles − voids = 0 (Section 28).

8 The Inner Direction and the Return

Because space is closed, every motion returns. This **return** does not run in space, but in an **inner direction** ϖ, orthogonal to space: a half-line from the place, closed at the **closure length** Λ. It is not a spatial direction. One cannot travel in it. It is the path on which the state of a place finds its way back to itself.

The return obeys a single equation, a random walk with loss, for the share u of the state that is at depth ϖ at time t:

$$\partial_t u + (\Gamma + \theta)\cdot u = D\cdot\partial_\varpi{}^2 u \qquad \Gamma = a_N/c \qquad D = GM/c$$

Here $a_N = GM/r^2$ is the Newtonian slope at the place, Γ the loss from the clock-rate gradient, θ the loss from the expansion of the region, and D the spreading in the inner direction.

Derivation of the return fraction. Stationary and without expansion, for the profile $u = G(\varpi)$ the equation $D\cdot G'' = \Gamma\cdot G$ remains, with $G(0) = 1$ (G here this profile, not the gravitational constant). The solution falls off exponentially, and at the return depth ϖ_R

$$q = G(\varpi_R) = e^{-\varpi_R\cdot\sqrt{(\Gamma/D)}} = e^{-s} \qquad s^2 = \Gamma\cdot\tau_D \qquad \tau_D = \varpi_R{}^2/D$$

q is the **return fraction**: the share of the state that comes back to the place.

The state space. A point of the state space is a complete state of a system, not a place and not a particle. Every independent direction of circulation is a degree of freedom, and n circulations span a torus T^n. Every force acts in it in one of three roles: **tangential** (it conserves energy and entropy and produces circulations and phases, like the electromagnetic one), **transverse** (it produces entropy and pulls onto the torus, like heat and radiation) or **parametric** (it slowly shifts the torus itself). In one equation: $\dot{z} = \mathscr{L}(z)\nabla E + \mathscr{M}(z)\nabla S + \varepsilon \cdot F_{par}$, with $\mathscr{L} = -\mathscr{L}^T$ and $\mathscr{L}\nabla S = 0$ (tangential), $\mathscr{M} = \mathscr{M}^T \geq 0$ and $\mathscr{M}\nabla E = 0$ (transverse). Gravity is parametric: it is not a direction of circulation, but fixes which orbits exist and which degrees of freedom are independent. It acts through two channels, the pre-tension (it changes the stiffness and with it which circulations exist) and the clock rate (it shifts all frequencies alike). A slow change keeps the action $J = E/\omega$ fixed, the torus moves to the neighbouring fibre of equal action. Every inner coupling term appears twice in the energy balance with opposite signs, it only shifts energy. Only gravity and externally driven parameters can supply it. A tangential coupling alone can never lock in. Locking needs loss.

Derivation of the threshold. An exactly open direction would have an infinite mean return time. The inner direction is closed at Λ. By Kac's theorem the mean return time to the place is then the path there and back, $\tau_D = 2\Lambda/c$, the same for every mass. With $\Gamma = a_N/c$ we get $s^2 = a_N \cdot \tau_D/c$, and with the **threshold**

$$a_0 = c/\tau_D = c^2/(2\Lambda)$$

it follows that $s^2 = a_N/a_0$. Kac's theorem holds for every closed state space: for every partition into regions, Σ_i $1/\langle T_{A_i}\rangle = 1$, the mean return time into a region being the total measure divided by its measure. Closure thus distributes a fixed budget of recurrence. An open space sets it to zero. For the dwell times of a closed, stationary system between two states, $\Sigma\ 1/\langle\tau\rangle = 1$: a direct test of closedness. If the region expands, the loss θ from the expansion is added. Whether the rule acts on a perturbation is then decided by the sum $s^2 = a_N/a_0 + \tilde{\theta}\cdot\tau_D$, not by a_N/a_0 alone, with $\tilde{\theta}$ the expansion rate of the region, the loss θ of the return equation. The model works with $a_0 = 1.06\cdot10^{-10}$ m/s^2. This belongs to $\Lambda = 13.7$ Gpc and $\tau_D = 89.6$ billion years. τ_D is longer than the age of the universe: the return is not a process along the orbit, but a stationary loop at a fixed place. Its depth follows from $\tau_D = \varpi_R^2/D$ with $D = GM/c$: $\varpi_R^2 = 2GM\Lambda/c^2 = r_0\cdot\Lambda$, and that is r^{*2} (Section 10). The return depth is the threshold radius.

9 The Rule

Counted at the place, all returns add up: the state comes back with the fraction q, of that again q, and so on. The renewal $u(0) = a_N + u(\varpi_R)$, each return an event in one sense of circulation, gives the **rule**:

$$a = \nu\cdot a_N \qquad \nu = \Sigma\, q^n = 1/(1-q) = 1 + n_B(s) \qquad n_B = 1/(e^s - 1) \qquad q = e^{-\sqrt{(a_N/a_0)}}$$

ν is called the **enhancement**. It has the form of one plus a Bose occupation. What is counted are returns, not particles. In general a feedback loop of gain q sums all returns to $\Sigma q^n = 1/(1-q)$: $q < 0$ damps, $0 < q < 1$ enhances, $q \geq 1$ runs away. The sign belongs to the whole closure, not to the single link.

The two limits. If the slope is large compared with the threshold ($a_N \gg a_0$), s is large, $q \to 0$, $\nu \to 1$: Newton holds exactly. If it is small ($a_N \ll a_0$), $\nu \approx 1/s$, so

$$a \to \sqrt{(a_N\cdot a_0)} = \sqrt{(GM\cdot a_0)}/r \qquad v^2 = a\cdot r = \sqrt{(GM\cdot a_0)} \qquad v^4 = GM\cdot a_0$$

The orbital speed becomes independent of the radius: this is the **flat rotation curve** of galaxies, and $v^4 \propto M$ is the Tully and Fisher relation. What the observer reads as dark matter is this changed rule of motion, not additional mass. The rule has exactly one branch: $d \ln a/d \ln a_N = 1 - \chi/2$ lies strictly between ½ and 1, without jump and without hysteresis, with the switch $\chi = s/(e^s - 1)$ (Section 15), and $d \ln a/d \ln a_0 = \chi/2$. The deep regime is scale-invariant ($r \to \lambda r$, $t \to \lambda t$, $a \to a/\lambda$). From this follows $v^4 \propto M$.

The transition lies at the **threshold radius**, where $a_N = a_0$:

$$r^* = \sqrt{(GM/a_0)}$$

There $\nu = 1.582$. **Example:** the Milky Way as a point mass of $6 \cdot 10^{10}$ solar masses has $r^* = 8.88$ kpc. The rule gives $v = 249$ km/s at 5 kpc, 214 at r^*, 196 at 15 kpc, 178 at 50 kpc, against the deep value of 170.4 km/s. The Sun has its threshold far outside, at $r^* = 7480$ AU. The planetary system lies deep in the Newtonian regime.

The balance. The rule is a balance like that between emission and absorption of light: emission $\propto 1 + n_B$, absorption $\propto n_B$, and their ratio is q. Newton is the spontaneous part, the return the stimulated one. Decomposed, $\nu = \frac{1}{2} + \frac{1}{2} \cdot \coth(s/2)$: a fixed half and the fluctuating part of an oscillator.

Light and orbits. The rule acts at the place of the orbit on all matter. The metric carries only its curl-free part, $\nu \cdot g_N = -\nabla\Phi + \nabla \times A$: light follows the metric and sees only $-\nabla\Phi$, while orbits also see the vortex $\nabla \times A$. The slope that light follows obeys $\nabla^2\Phi = \nabla \cdot [\nu \cdot \nabla\Phi_N]$, and both potentials of the metric are equal, $\Psi = \Phi$. For spheres the vortex vanishes, and there orbits and light read the same mass. In discs the dynamical mass inside is larger than the lensing mass. Far outside r^* the deflection of light tends to the fixed value $\alpha_\infty = 2\pi \cdot (v_{flat}/c)^2$, for the Milky Way 0.42″. Conversely, the flat rotation reads the fold directly, $\Delta s = v_{flat}^4/(3c^2 a_0)$. Except for spheres, the rule has no action in the sense of an energy: its force $-\nu\nabla\Phi_N$ is a gradient only when ν is constant on the Newtonian equipotential surfaces. It acts on the generator of motion (Section 29), not on the energy. On stellar orbits the exchange of work averages out, and the field form $\nabla \cdot [\nu\nabla\Phi_N]$ does not enter the rotation curves: the rule acts at the place of the orbit. In the solar system the return vanishes to every order. At the Earth's orbit the addition is e^{-7480}: the known tests of gravity remain untouched ($\gamma = \beta = 1$), and gravitational waves travel at c, with two polarisations transverse to the direction, from the changing quadrupole of the fold. Their sources always lie far above the threshold frequency $\Omega(r^*)/\pi$, so they are single funnels. In the strong field the return is without effect, waveform and ringdown are those of general relativity. The rise of the frequency, $\dot{f} = (96/5) \cdot \pi^{8/3} \cdot (3\Delta s_c/c)^{5/3} \cdot f^{11/3}$ with Δs_c the fold of the chirp mass of the pair, reads the fold of the source, just as the flat rotation reads that of a galaxy. The breathing is global and does not radiate.

10 The Triad and the Three Radii

Every field on the closed space splits into three parts, the **triad**: the **background** (harmonic, without sources, the loops), the **vortex** (rotation and remaining slope, the return at the place) and the **centre** (the flux out of the mass). The field of the parent level is background for the child level. In this way the triad repeats on every level.

Every body has, in addition, three radii:

$r_0 = 6\Delta s$ (centre) R (envelope, vortex) $r^* = \sqrt{(GM/a_0)} = c \cdot \sqrt{(3\Delta s/a_0)}$ (transition to the background)

The envelope R is the surface of the body. Between r_0 and R runs the vortex, outside r* the background carries. At the threshold radius the spread is the square of the flat rotation speed, $\delta(r^*) = 3\Delta s/r^* = (v_{flat}/c)^2$. Compact bodies have their threshold far outside their envelope and remain Newtonian. Galaxies have it within their extent. Atom, star, galaxy and cell are the same funnel. They differ only in two ratios: fold to envelope ($\Delta s/R$: clock rate and edge) and envelope to threshold (R/r^*: Newton or return). Dark matter is the second ratio, not a substance. The whole lies at its own threshold with r^*/Λ just below one.

Part III: Symmetries and Ladders

11 The Three Generators

The torus has not only loops, it has a **shape**: a flat torus can be square, rectangular or skewed. This shape is described by a single complex number, the **modulus** τ. Different values of τ can describe the same torus. The conversions between them are the symmetries of the torus shape. They form a group, $PSL(2,\mathbb{Z})$, and this group has exactly two **generators** from which every other symmetry is composed:

mirror	$\tau \to -1/\tau$	applied twice: the identity
threefold rotation	$\tau \to -1/(\tau + 1)$	applied three times: the identity
	$PSL(2,\mathbb{Z}) = \mathbb{Z}_2 * \mathbb{Z}_3$	

The sign $*$ is the free product: every symmetry is a chain of the two generators. To these belongs the **full turn** $\tau \to \tau + 1$: once all the way around, the closure per turn.

Their fixed points are the two distinguished shapes: the square torus (90°, fixed point of the mirror) and the hexagonal one (60°, 120°, fixed point of the threefold rotation). Only these rotations carry the shape of the torus into itself. This is why light that crosses the cell, the cube of the torus (Section 7), several times can only arrive rotated by 60°, 90°, 120° and 180°. The cell carries only the modes $k = 2\pi \cdot n/L$. At the best point of all precision data ($Q = 2\pi - 1$, maximum of the dark energy at saturation) L is 0.998 times the diameter of the last scattering surface, and the cube lowers the quadrupole of the background radiation by 13 % and raises the octupole by 2 %. On Planck's data below $\ell = 30$ this changes χ^2 by only 0.08, so the cell cannot be read yet.

Every symmetry of the shape is, by its **trace**, of one of three kinds:

Trace	Fixed point	Kind	In the model
below 2	inside	mirror, threefold rotation	**meeting points**
equal to 2	one, on the boundary, rational	full turn	**closure**, resonance
above 2	two, on the boundary, irrational	passing through, simplest: fixed point at the golden ratio	**open winding**

Meeting point, closure and open winding are the three kinds of one and the same group. Orbits run on tori with irrational winding, and no loop ever closes exactly. The golden winding is the most stable under perturbation, because it lies furthest from every rational resonance. Mirror, threefold and full turn, which recur on

every level of the model, are therefore not separate findings, but the whole symmetry of the torus shape.

Every phenomenon is thus the return of a closed path, counted at a place, weighted with a nome $q = e^{2\pi i\tau}$. Its modulus $|q|$ is the loss, with one sense of circulation: gravity, rule of return, weight of the closure rule, vacuum part. Its phase is the winding, lossless, in both senses of circulation: charge, spin, orbit, entanglement.

12 The Mirror on Every Level

A **mirror** is a fixed product of two quantities: if one grows, the other shrinks in the same ratio, and both lie symmetrically about their geometric mean. The basic case is the fold itself. Every mass has two lengths: the fold $3\Delta s = GM/c^2$, which grows with the mass, and the **quantum length** $\bar{\lambda} = \hbar/(Mc)$, which falls with the mass. Their product does not depend on the mass:

$$\bar{\lambda}\cdot 3\Delta s = (\hbar/Mc)\cdot(GM/c^2) = \hbar G/c^3 = \ell_P{}^2$$

The **Planck length** ℓ_P is the centre of the mirror. Both lengths meet at the Planck mass m_{Pl}. The coupling of gravity, $\alpha_G = (M/m_{Pl})^2$, is the distance from the meeting point.

The same form recurs on every level. Fanned out:

Level	Mirror pair	Fixed product (centre of the mirror)	Meeting point
every mass	fold ↔ quantum length	$\bar{\lambda}\cdot 3\Delta s = \ell_P{}^2$	Planck mass, $3\Delta s = \bar{\lambda} = \ell_P$
star	Planck length ↔ stellar fold	$\bar{\lambda}_p{}^2 = \ell_P\cdot 3\Delta s_*$	the holder reaches c (Section 21)
atom	electron radius ↔ Bohr radius	$\bar{\lambda}_e{}^2 = r_e\cdot a_B$	innermost electron at $Z\alpha = 1$
charge	electric ↔ magnetic charge	e·g fixed (Dirac)	Planck charge $q_P = e/\sqrt{\alpha}$
heat	thermal length ↔ quantum length	$\lambda_{th}\cdot\bar{\lambda} = 8\pi\ell_P{}^2$	Planck temperature
rotation	inner ↔ outer horizon	$a_J{}^2 = r_+\cdot r_-$	spin number j = 1
funnel	limit radius ↔ closure length	$r^{*2} = r_0\cdot\Lambda$	the whole, $r^*/\Lambda = 0.971$
part and whole	fold of the funnel ↔ fold of the whole	r* as geometric mean	the threshold of every funnel
return	mass ↔ mirror mass	$M\cdot\tilde{m} = m_{Pl}{}^2/2$	space ↔ inner direction
loops (all multiples)	Newton ↔ deep	$s \leftrightarrow 4\pi^2/s$	$s = 2\pi$, one full turn
whole and place	step integral ↔ local coefficient	$\zeta(k) \leftrightarrow \zeta(1-k)$	$(2\pi)^{-k}$, per radian
forces	Newton (r^{-2}) ↔ held body (r^1)	$(a+3)(A+3) = 4$	r^{-1}, the deep regime
vacuum	Planck density ↔ quantum density of the closure length	$\rho_s \propto \sqrt{(\text{Planck density}\cdot \hbar c/\ell_0{}^4)}$	$\varepsilon_L{}^2$ between 1 and $\varepsilon_L{}^4$

How each row arises.

- *Star:* with $3\Delta s_* = GM_*/c^2$, $M_* = m_{Pl}{}^3/m_p{}^2$ and $\ell_P\cdot m_{Pl} = \hbar/c$, one gets $\ell_P\cdot 3\Delta s_* = \hbar^2/(m_p{}^2c^2) = \bar{\lambda}_p{}^2$.
- *Atom:* the electron radius is $r_e = \alpha\cdot\bar{\lambda}_e$, the Bohr radius $a_B = \bar{\lambda}_e/\alpha$. Their product is $\bar{\lambda}_e{}^2$, and α drops out.

- *Charge:* by Dirac, the product of the electric charge e and the magnetic charge g is a whole multiple of the quantum of action. The meeting point of both is the Planck charge $q_P = e/\sqrt{\alpha}$.
- *Heat:* a fold has the temperature $kT = \hbar c/(24\pi\Delta s)$ (k the Boltzmann constant), its thermal length is $\lambda_{th} = \hbar c/kT = 24\pi\Delta s$. Multiplied by $\bar{\lambda} = \hbar/(Mc)$ this gives $8\pi\cdot\hbar G/c^3 = 8\pi\ell_P^2$.
- *Rotation:* a rotating funnel has two horizons $r\pm = GM/c^2 \pm \sqrt{((GM/c^2)^2 - a_J^2)}$ with the twist $a_J = J/(Mc)$, J the angular momentum. Their product is $r_+\cdot r_- = a_J^2$, their sum $6\Delta s$. At spin number $j = a_J/(3\Delta s) = 1$ both coincide. In the equatorial plane the clock rate of resting clocks remains exactly $\sqrt{(1 - 6\Delta s/r)}$. Co- and counter-rotating orbits differ by $4\pi\cdot a_J/c$, independent of the radius, and the twist separates the vertical cycle of an orbit from the revolution: the orbital plane precesses, the orbital torus goes from T^2 to T^3.
- *Funnel:* $r^{*2} = GM/a_0 = (2GM/c^2)\cdot(c^2/(2a_0)) = r_0\cdot\Lambda$.
- *Return:* Section 16. *Whole and place:* Section 15. *Vacuum:* Section 25. *Part and whole:* Section 23.
- *Loops:* if one counts not only the returns of a loop but all its multiples, the sum is $Z = \prod 1/(1 - q^n)$, up to $q^{1/24}$ the reciprocal of the η-function of the torus. Its mirror $\eta(-1/\tau) = \sqrt{(-i\tau)}\cdot\eta(\tau)$ maps the Newtonian regime onto the deep one, $s \leftrightarrow 4\pi^2/s$. In the deep regime $\ln Z = \pi^2/(6s) + \frac{1}{2}\cdot\ln(s/2\pi) - s/24$, and the leading part is the first step of the ladder.
- *Forces:* a force r^a and a force r^A are mirror partners when $(a + 3)(A + 3) = 4$: the map $u = z^{(a+3)/2}$ of the plane (z the place as a complex number) carries the orbits of one into the orbits of the other. With $u = z^2$, the ellipse about the centre, the orbit inside a uniformly held body (r^1), becomes the Kepler ellipse about the focus (r^{-2}). The mirror point of bound orbits is the exponent −1: exactly the force of the deep regime, $\sqrt{(GMa_0)}/r$. The second, the exponent −5, leads to the fall into the centre. The rule runs from Newton to the meeting point of the force laws.

Together they show: the levels of the world are mirror images of one another, reflected about fixed centres, and the whole stands to the part in the same form as the proton to the Planck point.

13 Meeting Points and Spread

The fixed points of the mirror are the **meeting points**: the places where motion reaches c. There, centre, mirror and envelope coincide. At the Planck point this happens completely ($3\Delta s = \bar{\lambda} = \ell_P$), and the funnel is degenerate. At the other end of the ladder the whole lies almost the same way, $r_0/\Lambda = 6/(1 + Q)$, $r^*/\Lambda = \sqrt{(6/(1 + Q))}$, with Q the weight of the mode of the whole (Section 24).

The **spread** of every level is the speed of its motion relative to c, the distance from the meeting point:

- For the atom it is $\alpha = v/c$. Winding is phase, an angle, linear in the speed.
- For the funnel it is $\delta(r^*) = 3\Delta s/r^* = (v_{flat}/c)^2$, the ratio of the radii $r_0/r^* = 2\delta(r^*)$. Fold is magnitude, a clock rate, quadratic like $\delta = v^2/c^2$.

Coupling constants are thus not arbitrary numbers, but distances from the meeting point of their level.

14 The Ladder of Lengths

Five lengths place every mass: the quantum length $\bar{\lambda}$ (at the place), the Planck length ℓ_P, the fold $3\Delta s$, the threshold radius r* (in space) and the **length of action** $\ell_0 = 2\Lambda$ (in the inner direction). They form a **ladder**. Where two lengths become equal lies a distinguished mass. Their positions hang on a single number, the ratio of the Planck length to the length of action:

$$\varepsilon_L = \ell_P/\ell_0 = 1.9\cdot10^{-62}$$

$M_{min} = \varepsilon_L\cdot m_{Pl} = \hbar/(c^2\tau_D)$	$\bar{\lambda} = \ell_0$
$M_c = \varepsilon_L{}^{1/3}\cdot m_{Pl} = 5.8\cdot10^{-29}$ kg	$r^* = \bar{\lambda}$
m_{Pl}	$3\Delta s = \bar{\lambda} = \ell_P$
$M_U = m_{Pl}/\varepsilon_L = c^3\tau_D/G = 1.1\cdot10^{54}$ kg	$r^* = 3\Delta s = \ell_0$

These are not bodies, but mass scales at which two lengths become equal. Both ends are set by the return time τ_D, the lower one through $\hbar$, the upper one through G. Without closure the ladder would have no ends. The mirror $M \leftrightarrow m_{Pl}^2/M$ exchanges $\bar{\lambda}$ and $3\Delta s$ and maps M_{min} onto M_U. The map $r \to r^{*2}/r$ is a mirror of the ladder as well: it maps the limit radius r_0 onto the closure length Λ, space onto the inner direction. In units of the upper rung M_U the Tully and Fisher relation is simply $M = M_U\cdot(v_{flat}/c)^4$, and the fold of this mass is $\Delta s(M_U) = \frac{2}{3}\cdot\Lambda$.

15 The Ladder of Steps

The function of the rule itself forms a ladder. Each step is the previous one times s:

Step d	$s^d\cdot n_B(s)$	Meaning	Integrated over all s
0	n_B	sum of returns	(grows without bound)
1	$\chi = s/(e^s - 1)$	switch from Newton (0) to the deep regime (1)	$\pi^2/6$
2	$s^2\cdot n_B$	additional acceleration, peak at $a_N = 2.54\cdot a_0$	$2\zeta(3)$
3	$s^3\cdot n_B$	structure action, vacuum part	$\pi^4/15$

At r* (s = 1) all steps are equal. Their peaks are Wien's displacement constants in d dimensions, $d(1 - e^{-s}) = s$. The deep regime is the Rayleigh and Jeans limit, the Newtonian regime the Wien tail: the rule has the same shape as black-body radiation, because both count returns within a period.

Whole and place. The same function has two readings. Integrated over all s, for the whole, each step gives $\Gamma(k)\zeta(k)$ with $k = d + 1$, $\Gamma(k) = (k - 1)!$ the gamma function (here not the loss Γ) and ζ Riemann's zeta function, $\zeta(k) = \Sigma\ 1/n^k$. At the place, expanded near the deep regime, it has the coefficients $\zeta(0) = -\frac{1}{2}$, $\zeta(-1) = -1/12$, $\zeta(-3) = 1/120$:

$$n_B(s) = 1/s + \Sigma\,\zeta(-n)\cdot(-s)^n/n! \qquad \zeta(1 - k) = 2\cdot(2\pi)^{-k}\cdot\cos(\pi k/2)\cdot\Gamma(k)\zeta(k)$$

Riemann's mirror connects the two. Thus the constant of the vacuum part, $\Gamma(4)\zeta(4) = \pi^4/15$, is the mirror image of $\zeta(-3) = 1/120$, which the observer at the place reads as the correction $-(r^*/r)^3/720$, and $\pi^2/6$ is the mirror image of $-1/12$. What carries the whole and what appears at the place as a correction are the same quantity seen from two sides.

16 The Return as a Mirror Image

If one reads the time of the return as imaginary time, the return equation becomes the Schrödinger equation in the inner direction ϖ, with $D = \hbar/(2\tilde{m})$ and the slope as the potential, $V = \hbar\Gamma = \hbar a_N/c$.

Derivation. From $D = GM/c$ and $D = \hbar/(2\tilde{m})$ follows the **mirror mass**

$$\tilde{m} = \hbar c/(2GM) = m_{Pl}^2/(2M) \qquad \bar{\lambda}(\tilde{m}) = \hbar/(\tilde{m}c) = 2GM/c^2 = r_0$$

The funnel M returns like the quantum motion of its mirror mass, and the quantum length of that mass is its limit radius. The return fraction $q = e^{-s}$ is the tunnelling factor of this mirror image through the slope, the same form as $e^{-\sqrt{(E_G/E)}}$ when two nuclei of energy E pass through their Coulomb barrier (E_G the Gamow energy). This is why the period of the return is a temperature, $kT_I = \hbar/\tau_D = \hbar a_0/c$, and why the rule has the Bose form: heat is periodicity in imaginary time. The geometric distribution of the number of returns has, at fixed mean, the largest entropy, $H = \ln \nu + \chi$. The threshold itself compares two such quantities: the Unruh temperature T_U of the observer held in place against the slope, with the period of the inner direction, $s^2 = 2\pi \cdot T_U/T_I$. This is a period, not heat at the place. Gravity and the quantum are two views of the same mirror, M in the spatial direction, m_{Pl}^2/M in the inner one.

The meeting point of the ladder of steps and the mirror point of the loops also count the same loop: with $s = -\ln|q| = 2\pi \cdot \mathrm{Im}\ \tau$, the meeting point $s = 1$ lies at one radian, the mirror point $s = 2\pi$ at the full turn. The threshold counts per radian, the mirror per turn.

Part IV: The Levels

The same formulas hold on every level. Only mass and radius change. Every level receives its position from its neighbours: the end of the nucleus from the coupling α of the shell, the window of matter from the kind of bond of the atom, the spread of the star from what holds it, the burning stages from the iron of the nucleus, the smallest gas-bearing galaxy from the heat of its gas, the position of the galaxy relative to the threshold from the whole.

17 Particle, Coupling, Charge

An elementary particle is an oscillating state of the medium. Next to it, space remains practically flat. Gravity and electromagnetism have the same form, a **length of action divided by the quantum length**:

$$\alpha_G = 3\Delta s_p/\bar{\lambda}_p = Gm_p^2/(\hbar c) = 5.906 \cdot 10^{-39} \qquad \alpha = r_e/\bar{\lambda}_e = 1/137.036$$

Two ways of imprinting space at a node must be distinguished. The **fold** is a magnitude: it changes the clock rate and has one sense of circulation, that of loss. The **winding** is a phase: it does not change the clock rate, is lossless and counts in both senses of circulation. Mass is fold, **charge is winding**: the phase of a node has a whole-number winding n around its circulation, the charge is $Q_{el} = n \cdot e$ with e the elementary charge, and the sign is the handedness. On the torus, n is a topological number (the first Chern number). It changes only when the amplitude passes through zero. This is why Coulomb's law remains $1/r^2$ in every field, and why there is no electromagnetic threshold effect: the argument of the rule of return is the clock-rate gradient, never an elec-

tric force. In the state space the Lorentz force acts tangentially: on a moving charge its magnetic part does no work, it conserves energy and entropy and produces circulations and phases. Joule heat $J \cdot E$ is the channel into the transverse role. By Onsager and Casimir the transport splits into a symmetric part (conductivity, heat conduction, Seebeck, Peltier: transverse) and an antisymmetric one (Hall, Nernst: tangential). The magnetic field decides which share circulates and which is lost.

Twist and three rates. Fold and winding sit at the same node. The twist $a_J = J/(Mc)$ is an imaginary displacement of this place. Both multipole series are $M \cdot (i \cdot a_J)^l$ and $Q_{el} \cdot (i \cdot a_J)^l$ (i the imaginary unit, l the order), hence $J = Mc \cdot a_J$ and $\mu = Q_{el} \cdot c \cdot a_J$: from this follows, for every mass, charge and rotation, the same ratio of magnetic moment to angular momentum, $g = 2$. For the electron this is the Bohr magneton $\mu_B = e\hbar/(2m_e)$. Every mass has three rates: the fold rate $c/(3\Delta s)$, the Compton rate $c/\bar{\lambda}$ and the folding rate $\alpha_G \cdot c/\bar{\lambda}$. The Compton rate is their geometric mean, once more a mirror. Below the Planck mass the fold rate is the fastest, above it the reverse, at the Planck mass all three are equal.

Return of the winding. In the field B the twist turns at the rate eB/M, for $g = 2$ exactly as fast as the orbit goes round, the orbit as a whole only half as fast. The difference the measurement reads is made by the return of the winding onto itself: counted per radian, with the weight α, once back, $\mu/\mu_B = 1 + \alpha/(2\pi) = 1.001161$, measured 1.001160 (CODATA 2018). Independently, the hydrogen line, winding times twist at the place of the nucleus, reads the first return as $0.88 \cdot \alpha/2\pi$: without return 1418.95 MHz, with the first 1420.60 MHz, measured 1420.41 MHz. When the rate eB/M reaches the Compton rate, the field lies at the meeting point of the winding, $B = M^2c^2/(e\hbar) = 4.41 \cdot 10^9$ T (Schwinger field): there pairs arise from the medium.

Matter in the field. The three roles give three responses. Tangentially every orbit turns against the field without loss, the diamagnetism of every substance (hydrogen atom $\chi = -2.38 \cdot 10^{-6}$ cm^3/mol). Open windings align against the loss kT, paramagnetism, with two senses of circulation in the Fermi form $m = \tanh(\mu B/kT)$, small the Curie law. If the windings of neighbouring atoms lock in the same sense, the field of the others is a feedback with $q = T_C/T$: $\chi = (C/T) \cdot \Sigma q^n = C/(T - T_C)$, the tipping point $q = 1$ is the Curie temperature, ferromagnetism. Locking in opposite senses is $q < 0$, the antiferromagnet. Fully locked is the superconductor: two locked windings obey $\dot{\psi} = \Delta\omega - K \cdot \sin\psi$, the equation of the locking of two cycles (Josephson), the phase of the pair closes per circuit with $\Phi_0 = h/(2e) = 2.068 \cdot 10^{-15}$ Wb, and the transverse channel is closed: no loss, no resistance, the field penetrates only to the closing length λ_L (Meissner).

18 The Nucleus: a Held Funnel

A **held funnel** is a body whose building blocks are held at the distance of a **holding length** λ_H. Its envelope is then $R = \lambda_H \cdot N^{1/3}$ for N building blocks. The atomic nucleus is such a funnel, held by its nucleons themselves.

Derivation of the holding length from nuclear masses. Through the mutual repulsion of its protons, the binding of a nucleus with Z protons and mass number A loses the Coulomb energy $a_C \cdot Z(Z-1)/A^{1/3}$ with $a_C = (3/5) \cdot \alpha\hbar c/\lambda_H$. Fitting the liquid-drop formula to all nuclei of AME2020 gives $\lambda_H = 1.25$ fm. Independently of that, **mirror nuclei**, in which Z and N are exchanged, differ only in winding. Their difference in binding is pure Coulomb energy and reads the holder as $\lambda_H = 1.33$ fm (36 pairs, AME2020).

The nucleons move inside the nucleus with $v/c \approx \bar{\lambda}_p/\lambda_H = 0.17$. Half the spread of the binding of ^{62}Ni independently gives 0.14. Size and binding read the same speed up to a factor of order one. Nuclear transitions run by themselves towards the smallest fold per nucleon, at ^{56}Fe. The largest binding per nucleon belongs to ^{62}Ni (AME2020).

The tipping point. The fissility $x = E_C/(2E_S)$ compares the Coulomb energy E_C of the protons (winding) with the surface energy E_S of the holding. It is the gain of a feedback that tips over at $x = 1$: the restoring force of a deformation is $\propto 1 - x$. The binding per nucleon is largest at $x = ¼$ (fitted to the measured nuclei of AME2020: $x = 0.25$). The tipping point is the meeting point of the nucleus. The heaviest known nuclei lie there ($Z = 118$, $A = 294$: $x = 0.98$), while the innermost electron of the same atoms is only at $Z\alpha = 0.86$: matter ends at the meeting point of the nucleus before the shell reaches its own. Because x grows with α, the coupling of the shell sets the end of the nucleus.

The valley. The heaviest known nuclei are poor in neutrons. In the valley of stability, where at fixed A the fold energy $Z\cdot E_H + N\cdot E_n - B$ is smallest (E_H, E_n the rest energies of hydrogen atom and neutron, N the number of neutrons, B the binding) (for each A the continuous Z of smallest fold energy meets the lightest measured isobar to within 0.32 protons, 219 mass numbers up to $A = 238$), the tipping point by the liquid-drop formula lies only at $Z = 134$ ($A = 370$), with the shell term of the model's closure numbers at $Z = 181$ to 191. The formula with shell term fits the data better, the scatter of the binding falls from 3.39 to 1.72 MeV. The tipping point itself is not measured, masses only up to $Z = 110$. The shell holds up to $Z = 170$: $Z\alpha = 1$ ($Z = 137$) holds for a point-like nucleus. Around the nucleus with $R = \lambda_H\cdot A^{1/3}$ the binding of the innermost electron reaches twice the fold energy of the electron, $6c^4\cdot\Delta s_e/G$, only there (Dirac equation).

The shells. The shell term follows from the funnel itself. The levels of the held funnel with twist ($\kappa = 0.12$, $\mu = 0.03$) close at 2, 8, 20, 28, 50, 82 and bind against their averaged filling with the shell spacing $\hbar\omega = (5/6)\cdot(3/2)^{4/3}\cdot\hbar^2/(m_N\cdot\lambda_H{}^2)\cdot A^{-1/3} = 38.0\ \text{MeV}\cdot A^{-1/3}$. AME2020 reads independently 39.3, hence $\lambda_H = 1.23$ fm. With no free number in the shell term the binding scatters by 1.83 MeV, the spherical nuclei lack the closure at 126. At the ground-state spins (NUBASE2020, 850 odd nuclei with unambiguously measured spin) the filling meets 75 % of the parities and 38 % of the spins, outside the deformed regions 86 % and 49 %: the parity carries, the spin half of it.

19 The Atom, Closing and Opening

An atom is a small, fast-circulating vortex: tiny fold, large steepness. The shell of hydrogen circulates with $v/c = \alpha$ on the Bohr radius $a_B = 0.529\cdot10^{-10}$ m, at $6.58\cdot10^{15}$ Hz. The atom too has its three radii: the threshold radius of the proton lies at $r^* = 32$ fm, between nucleus and shell ($a_B = 1631\cdot r^*$). As a compact funnel, its interior remains Newtonian. The binding of the atom comes from the geometry of its shell, not from the depth of its fold.

Closing forms. The pure $1/r^2$ slope closes every orbit, and orbits of equal energy bind equally, whatever their shape. This has a reason in form: the momenta of a Kepler orbit lie on a circle, and mapped onto the sphere S^3 every orbit is a great circle. In the space of motions, not in position space, the funnel is a closed sphere. The n^2 states of a shell (n its number) are the vibrations of this sphere, $2n^2$ with both senses of circulation. Only two slopes close every orbit (Bertrand's theorem): the $1/r^2$ slope outside a funnel and the linear one inside a uniformly held body. Their shells close after

funnel ($1/r^2$):	2, 10, 28, 60	$2n^2$ per shell
held body (r^1):	2, 8, 20, 40, 70, 112	$O(k) = (k + 1)(k + 2)(k + 3)/3$

Opening couplings. Every deviation from the pure form opens the closure, on every level: in the solar system the fold (3δ per revolution), in the galaxy the rule and the extended mass (rosettes, in the deep regime 254.6° lie between the pericentres), in the atom the inner shells that spread out the centre, in the nucleus the

twist. Shell and nucleus open in mirrored fashion:

- In the shell, the orbits with the **smallest** orbital winding l bind more strongly, because they dive deep. Winding here means the number of nodes of the oscillation around the centre, not the winding of the charge (Section 17). The shells fill by n + l and close after 2, 10, 18, 36, 54, 86 instead of after 28 and 60.
- In the nucleus, the twist pulls down, in every shell, the state with the **largest** orbital winding, j = l + ½ (j here the angular momentum of the nucleon, not the spin number), from the next shell. The nucleus then closes after O(k) + 2(k + 2), with k the number of the shell, that is after 28, 50, 82, 126 (k = 2 to 5). The next closure would be 184.

The orbit in a galaxy has a torus too. Its modulus τ_M carries in its real part the ratio of radial to azimuthal frequency, in its imaginary part the binding. For the point mass $(\mathrm{Re}\ \tau_M)^2 = 1 + \chi$: $\chi = 0$ gives the closed Kepler orbits, $\chi = 1$ rosettes. The rule lifts the Kepler degeneracy $\kappa = \Omega$ (Ω the frequency of the orbit, κ that of the radial oscillation about the circular orbit) and replaces it by local resonances at rational κ/Ω.

In the data. The six strongest drops of the ionisation energy lie after Z = 2, 10, 18, 36, 54, 86 (factor 4.6 at He down to 2.8 at Rn, PubChem), the six strongest jumps of the two-nucleon binding at N = 16, 20, 28, 50, 82, 126 and Z = 14, 20, 28, 38, 50, 82 (AME2020). The series meets 20, 28, 50, 82 and 126. The jumps at N = 16 and Z = 14, 38 lie at subshells within the series, and 8 does not show in the two-nucleon binding. The closure numbers of both levels thus follow from one closing form each and one coupling that opens it.

The ladder of periods. With the noble gas Og the seventh period closes at Z = 118 (PubChem). The periods are 2, 8, 8, 18, 18, 32, 32 long. The length of period p is the number of states with n + l = p + 1, that is $2k^2$ with k the rounded-up half of p + 1: every step $2k^2$ of the $1/r^2$ form comes twice, because n + l = 2k − 1 and 2k have the same largest winding l = k − 1. The ladder is itself a closing form: a field that is mirrored in log r about a radius R ($r \leftrightarrow R^2/r$) binds at the threshold E = 0 exactly at the couplings N(N + 1) with N = n + l, each step with $2k^2$ states, k the rounded-up half of N. Its states at the threshold are spherical harmonics, $u = \sqrt{r}\cdot P_N^{2l+1}((r - R)/(r + R))$ with P the associated Legendre function: the step n + l is the degree on a sphere whose poles are nucleus and infinity and whose equator is R. The winding l gives the order 2l + 1, and the mirror $r \leftrightarrow R^2/r$ is the reflection at the equator. In log r this field is $v\cdot f(1 - f)$ with $f = 1/(e^x + 1)$, $x = \ln(r/R)$: the same function as the rule, only in the other counting, the winding in both senses of circulation, $f(x) + f(-x) = 1$, instead of the magnitude in one, $\nu(s) + \nu(-s) = 1$. The shell is a funnel held by the counting rule of the electrons, like the white dwarf, with $R \propto Z^{-1/3}$ (Thomas and Fermi): its field follows the mirror form inside to 0.6 % and opens outside. Its height $v = 3.44\cdot Z^{2/3}$ lies from Ne to Og in the window $N(N + 1) < v < (N + 1)(N + 2)$ with N = p + 1, the field binds exactly the occupied steps. 99 of the 118 elements fill exactly by n + l. The 19 deviations lie in d and f shells within the periods and leave the closure numbers untouched: eight of them close a half or full shell (d^5, d^{10}, f^7), seven begin the f shell with a d electron, the smaller winding, four give an s electron to the d shell.

Beyond 118. The eighth period would have 50 places, with 8s (Z = 119, 120) and the first g shell (l = 4), and would close at 168. There the height of the held funnel lies in the window (104.8 between 90 and 110). Held in the valley by both formulas are the elements up to Z = 133, only close to the valley (N ≈ 202 to 235). By the formula with shell term they reach the limit of the shell, Z = 170. Alpha decay is a passage through the Coulomb threshold, $\lg T_\alpha = a\cdot Z/\sqrt{Q_\alpha} + b\cdot Z + c$ with T_α the half-life, Q_α the energy of the decay and three fitted numbers a, b, c. On 82 even-even alpha emitters (NUBASE2020, AME2020) this fits to 0.51 decades, and in the held range the nuclei in the valley would hold against alpha decay for seconds to hours.

20 Matter: the Coupling Window

Two coupled cycles with phase difference ψ obey $\dot\psi = \Delta\omega - 2k\cdot\cos\varphi\cdot\sin\psi$, with $\Delta\omega$ the difference of their eigenfrequencies, the coupling k (here not the Boltzmann constant) and its phase φ. They lock in when $|\Delta\omega| \le 2k|\cos\varphi|$, and their torus then contracts onto a circle. Only the lossy part $k\cdot\cos\varphi$ of the coupling locks, the rest merely shifts the frequencies. Too weak a coupling lets the cycles drift apart. In general, for cycles θ and slow structure Z: $\dot\theta = \omega(Z) + K(\theta, Z)$, $\dot Z = G(\theta, Z) - D(Z)\cdot Z$, with eigenfrequencies ω, cross-coupling K, feed G and loss D. A common medium with pair coupling k carries at most $N \approx 1 + 2W/(\pi k)$ free cycles in a frequency band W. Above that they lock in. If a feedback grows to $q = 1$, Σq^n diverges: before this tipping point $1 - q \propto \delta^{1/2}$ (δ here the distance to the tipping point), the hysteresis window opens with the distance to the power 3/2, and the return rate falls with its square root (critical slowing down). From T^2 to T^1 lead three ways: locking (both cycles go on oscillating, bound), kinking, and the loss of a cycle. The number of independent cycles is the number of neutral Lyapunov exponents, transverse to the torus they are negative. Stability is motion without an obligation to repeat: the feedback must take hold before a perturbation builds up. Bifurcations are changes of role: the birth of a cycle makes a transverse direction tangential, locking makes a tangential one transverse. Hysteresis needs no store, the state is only the sequence of the extremes not yet erased. Noise and loss come from the same damping. In between lies the **coupling window** $k_{min} < k < k_{lock}$. A substance is the coupling window of its atoms: solid is locked (the lattice as identification), liquid is inside the window, gaseous is frayed. Melting gives up the identification of the lattice, not the cohesion.

The width of the window depends on the level below: across 93 elements, T_{boil}/T_{melt} ranges from 0.81 (As, which sublimes, so the window is closed) to 8.2 (Ga), and the more firmly the atom binds, the narrower the window (rank correlation with the ionisation energy −0.48, PubChem). The trend runs between kinds of bond (noble gases and halogens narrow, metals wide), not within a group.

21 Stars

Mass scale. The fold of a star is fixed by the fold and the quantum length of the proton:

$$M_* = m_{Pl}^3/m_p^2 = 1.853\ M\odot \qquad \bar\lambda_p^2 = \ell_P\cdot 3\Delta s_*$$

The star is the mirror image of the Planck point at the quantum length of the proton: its fold stands to the quantum length of the proton as the latter stands to the Planck length.

Held stars. What holds the star sets its spread. If the $N = M/m_p$ nucleons are held at the distance of a holding length λ_H, then $R = \lambda_H\cdot N^{1/3}$, and on the mass scale M_* the spread is

$$\delta = \bar\lambda_p/\lambda_H$$

the ratio of the quantum length of the carrier of the fold to that of the holder (up to factors of order one). Three cases follow from this one form:

- If heat holds it ($\lambda_H = \hbar c/kT$), this is the central temperature of the star, $kT_c = f\cdot\delta\cdot\bar m c^2$ with $\bar m$ the mean mass of the particles and a factor f of order one.
- If the **counting rule** of the electrons holds it, the rule that every state holds only one winding per sense of circulation (the closure of the shells, Section 19), at the distance $\bar\lambda_e$, then $\delta = m_e/m_p$: the white dwarf. The same balance, the pressure of the counting rule against the slope, gives the limiting mass M =

$\omega_3\cdot(\sqrt{(3\pi)}/2)\cdot m_{Pl}^3/(2m_u)^2 = 1.456\ M\odot$ ($\omega_3 = 2.018$ the coefficient of the form of the funnel, two nucleons per electron, m_u the mass unit).

- If the counting rule of the nucleons themselves holds it, δ is of order one: centre and envelope almost coincide, the meeting point at which the holder reaches c.

The burning stages lie as a ladder in the depth as long as the binding per nucleon grows, up to $x = \frac{1}{4}$: in the valley by the liquid-drop formula $Z = 26.7$, between ^{56}Fe and ^{62}Ni (AME2020). If the counting rule of the electrons holds the funnel, it reaches at most $kT = 0.096\cdot(\mu/\mu_e)\cdot m_{ec}^2$ (μ and μ_e the mean mass per particle and per electron in units of m_u), for carbon and oxygen $5.0\cdot10^8$ K. Beyond the iron group only the capture of neutrons leads on. That the star on the mass scale lies at the border between radiative and convective envelopes is the condition $\alpha_G \approx \alpha^{12}\cdot(m_e/m_p)^4$ (Carter 1974: pressure balance, ignition, radiative transport with the Eddington limit $L_{Edd} = (9/2)\cdot m_p\cdot c^3\cdot\Delta s/r_e^2$ and emission together), with the numbers $2.0\cdot10^{-39}$ against $5.9\cdot10^{-39}$, equal up to a factor of order one. Gravity acts here only collectively, through $\alpha_G^{-3/2}$ nucleon folds.

The fold. When the shell falls onto the centre, $R = r_0 = 6\Delta s$, the funnel becomes a **fold**, the black hole. Its depth remains the finite fold Δs. Only someone who takes the depth of the funnel for a place reads a singularity. Of the three radii two remain, r_0 and r^*, with $r^{*2} = r_0\cdot\Lambda$: mirrored at the threshold radius, the horizon is the closure length. At the horizon $s^2 = M_U/(4M)$. The fold lies deep in the Newtonian regime there, and inside a galaxy it remains a single funnel ($10^6\ M\odot$: $s^2 = 1.4\cdot10^{17}$ at the horizon, $r^* = 36$ pc). Only at $M_U/4 = 1.4\cdot10^{23}\ M\odot$ do horizon, threshold and closure length coincide, the meeting point that the whole almost reaches.

The fold is its own mirror: its horizon is the quantum length of its mirror mass, $r_0 = \bar{\lambda}(\tilde{m})$, and the horizon of the mirror mass is its own quantum length, $r_0(\tilde{m}) = \bar{\lambda}(M)$. Its temperature is the rest energy of the mirror mass divided by 4π, $kT = \tilde{m}c^2/(4\pi) = \hbar c/(24\pi\Delta s)$. Against the period of the return, $kT_I = \hbar a_0/c$, $T/T_I = M_U/(8\pi M)$: the two are equal when the horizon is the closure length per radian, $r_0 = \Lambda/2\pi$, the same counting as for the threshold. One solar mass has a period $2.3\cdot10^{22}$ times shorter. What falls into it releases, down to the innermost stable orbit ($18\Delta s$, $\delta = 1/6$), $1 - (1 - 2\delta)/\sqrt{(1 - 3\delta)} = 1 - \frac{2}{3}\sqrt{2} = 5.7$ % of its rest energy, six times as much as nuclear binding up to iron. At the meeting point of rotation ($j = 1$) it is $1 - 1/\sqrt{3} = 42$ %, and the fold is cold there because both horizons coincide.

22 Galaxies: the Rule in the Data

A galaxy is a collective funnel: stars are **single funnels**, while the stellar system as a whole follows the rule in its total field. Compact funnels, denser than the tidal density of their surroundings, remain Newtonian up to their smoothing length. Only the collective field, smoothed over them, is enhanced. The tidal density is $\rho_J = 3\mathscr{T}/(4\pi G)$ with the tide $\mathscr{T}$ of the smoothed parent field, and the smoothing length ℓ follows from $\ell^3 = GM/\mathscr{T}$. Because the rule acts at the place on the smoothed field, in which the body's own field vanishes at its centre of mass, bodies at the same place fall alike. Compact globular clusters ($g = 25\cdot a_0$) show hardly any addition ($\nu = 1.007$), thin systems ($g = 0.13\cdot a_0$) the full one ($\nu = 3.3$). The star clusters of the Local Volume Database (52 with measured dispersion, g_N at the half-light radius with $Y_V = 2$) show this: with $Y_V = 0.93$, calibrated on the 28 dense ones ($g_N > 3\cdot a_0$), the 17 thin ones ($g_N < a_0$) lie $+0.10 \pm 0.03$ dex above the prediction of the rule and +0.41 above Newton, as in the standard model without a halo (χ^2 53 against 235). The 11 well measured ones lie +0.03 against +0.34, the 9 of the Milky Way farther than 10 kpc from its centre +0.06 against +0.43. The residual against Newton depends more on g_N (rank correlation −0.51 over 14 clusters of the Milky Way) than on the tide, which at their present distance fills their Jacobi radius to 0.13 to 0.63

(+0.31). Pressure-supported spheres have, isotropically and for every profile, $\sigma_v^4 = (4/81)\cdot GMa_0$ (σ_v the speed of their thermal motion): 9.1 km/s at 10^7 M☉, 19.3 km/s at $2\cdot10^8$ M☉. As a whole, a system lies below the threshold with surface density $\Sigma < a_0/(\pi G) = 242$ M☉/pc^2.

Surroundings and dimension. The rule acts on the total field of a region's own masses, not particle by particle. If the region lies in the field of a larger one, it falls with it: the uniform field cancels in the fall, and inside, the region reads only its own folding. What stays effective is the tide, and it decides whether the region remains collective. As a whole it moves in the enhanced field of its host. At the same time the rule lowers the dimension of every form: Newton is flux through the spherical surface, the deep regime flux through a circumference, $d_{eff} = 1 + (d - 1)(1 - \chi/2)$. The sphere goes from 3 to 2, a filament from 2 to 1.5, a wall stays at 1. A filament of line mass μ has the threshold $r^*_{fil} = 2G\mu/a_0$ (at 10^{12} M☉/Mpc: 2.6 kpc). Deep inside it the circular speed grows like $r^{1/4}$. The epicycles change as well: instead of $\kappa^2 + \nu_z^2 = 2\Omega^2$ (ν_z the frequency of the oscillation perpendicular to the disc), $(2 + \chi)\Omega^2$ holds. The observer who assumes unchanged rules reads the missing part as a density, $4\pi G\rho_X = \chi\Omega^2$, outside an isothermal halo.

Surroundings in the data. The data carry this twice. Around isolated galaxies (KiDS-1000, Brouwer 2021) the lensing of the blue ones, which still form stars, lies on the rule without external field ($\chi^2 = 21$ in 15 bins), already $0.003\cdot a_0$ in magnitude raises it to 534. The 22 classical satellites of the Local Group, whose host field exceeds their own, lie $+0.15 \pm 0.04$ dex above the rule without the host field, with it $+0.46 \pm 0.05$ ($\chi^2 = 76$ against 341).

In the data (SPARC). On the rotation curves of 141 galaxies with 2688 points, with distance, inclination and mass-to-light ratio estimated for each galaxy, the threshold reads $a_0 = 1.11\cdot10^{-10}$ m/s^2 (16 to 84 %: 1.00 to 1.21). The mass-to-light ratio of the discs follows from the requirement that stars and gas obey the same rule, because the rule acts on the fold, not on the state of matter: $Y_d = 0.67$. With the value of the stellar models, $Y_d = 0.5$, one would read a higher threshold and an apparent trend from stars to gas, two consequences of the same foreign conversion. The central surface density from the rotation curve lies, galaxy by galaxy with its own baryons, on average 0.05 dex below the rule, compatible. The Bose form fits best. The simple form, $\nu = ½ + \sqrt{(¼ + a_0/a_N)}$, and the standard form, $\nu = \sqrt{(½ + \sqrt{(¼ + (a_0/a_N)^2)})}$, are worse by $\Delta\chi^2 = 53$ and 41 (χ^2 the error measure of the fit, not the switch χ). The return has dimension 0.92 ± 0.05, one-dimensional as the square root in q requires. The residual per galaxy depends neither on mass nor on gas fraction, neither on shape (Hubble type) nor on surface brightness: the rule reads only the fold.

Against the halo. For the same curves the standard model needs an NFW halo with two free numbers per galaxy. The rule has $\chi^2 = 2918$, the halos 1434 with 282 more free numbers. By the information measure BIC (χ^2 plus the number of free numbers times ln 2688) the rule leads by 736, 2926 against 3661. For 31 of the 141 galaxies the halos lie at the edge of their range, and their mass scatters against that of the baryons by 0.58 dex: the halo is bookkeeping, not an object of its own.

Discs at the threshold. The disc scale length R_d is, in the median, $0.71\cdot r^*$ (16 to 84 %: 0.37 to 1.28) across four decades of mass. That is $r^*/\sqrt{2}$: the centre of the disc lies exactly at the threshold of surface density, $\Sigma_0 = M/(2\pi R_d^2) = a_0/(\pi G)$. It grows with the baryonic mass as $M_b^{0.31 \pm 0.02}$: the envelope as at fixed density of the whole with $M^{1/3}$, the threshold from the galaxy's own fold with $M^{1/2}$, hence $R/r^* \propto M^{-1/6}$, like R_{ta}/r^* in the inactive measure (Section 26).

Memory of binding. The return of a bound region reads the baryons it has bound, the slope at the place only those in it today: $a = a_N + (\nu(a_N,B) - 1)\cdot a_N,B$, with a_N,B from the fold of those once bound. This is the decomposition of the rule after Einstein, $\nu = 1 + n_B$: Newton, the spontaneous part, reads at the place what is

there today, the return n_B, the stimulated part, in the inner direction what was bound. With today's gas and $Y_V = 2$ the gas-rich isolated dwarfs lie 0.00 ± 0.04 dex on the rule, the gas-poor isolated +0.13 ± 0.07, the 36 classical satellites +0.11 ± 0.03, the ultra-faint +0.37 dex (Local Volume Database), and the tide does not order this. Two independent routes read the gas they once carried. The gas of isolated dwarfs of the same luminosity (188 dwarfs) puts the classical satellites +0.011 ± 0.028 dex on the rule. The metallicity of their stars reads, in the closed box, $\langle Z\rangle = p\cdot(1 + \mu\cdot\ln \mu/(1 - \mu))$, the gas fraction μ at quenching, with the yield from the gas-rich ones ($\lg p = -1.03$): then the classical lie −0.03 ± 0.03 dex on the rule, the ultra-faint +0.07 ± 0.04, the gas-poor isolated +0.05 ± 0.08. Discs that still form stars still carry their gas: in SPARC the residual depends neither on the gas fraction nor on the type. Around isolated galaxies the blue lenses lie +0.01 ± 0.02 dex on the rule, the red ones inside +0.13 ± 0.02 above it.

What the observer sees. Whoever assumes unchanged rules converts the addition into a mass:

$$M_{apparent}/M = 1/(1 - e^{-s}) = r/r^* + \tfrac{1}{2} + r^*/(12r) - (r^*/r)^3/720 + \dots$$

The fixed half ½ and the coefficients 1/12 and 1/720 are exactly the local coefficients of the ladder of steps. They are Bernoulli numbers: the error of the single returns against the continuum, as of the polygon against the circle. The additional potential that the observer ascribes to an invisible mass is $\Phi_X = v_{flat}^2\cdot\ln \nu(s)$, the logarithm of the mean number of passes. The density that is read off is an output of the observer's calculation, not a content of space. Read as mass, the share of the baryons in it changes sign at the largest additional acceleration: thin matter enhances it, dense matter reduces it.

Part V: The Whole

23 The Threshold from the Whole

The threshold $a_0 = c^2/(2\Lambda)$ is the same for every funnel. Where does Λ come from? If one reads the closure length as the entire fold of the cell, counted with its returns,

$$\Lambda = (1 + Q)\cdot\Sigma\Delta s_b(\text{cell}) \quad \Rightarrow \quad a_0 = 3c^4/(2G\cdot\rho_m\cdot L^3) \quad L = 27.9\ \text{Gpc}$$

with ρ_m the mean density of matter, baryons and mode, then the threshold is tied to the spatial cell L of the torus. Today this is the diameter of the last scattering surface. Read this way, Λ is the fold of the whole per direction, and the threshold becomes a mirror between part and whole: in $r^{*2} = r_0\cdot\Lambda$, r^* is the geometric mean of the fold length $r_0 = 2GM/c^2$ of the single funnel and the fold of the whole, just as ℓ_P is the geometric mean of $\bar{\lambda}$ and $3\Delta s$. **Every funnel reads its threshold from the whole.** This reading of Λ is posited. The mirror between part and whole supports it.

24 The Closure Rule and Q

In the background radiation and in the build-up of clusters, the local rule is not enough. There a restriction of the degrees of freedom of the whole acts, the **closure rule**, not a substance:

$(\partial\xi)^2 = -1$	the pressure direction of the conformal mode is closed
$\dot{z} = (I - w\cdot\Pi\perp)F$	$\Pi\perp$ projects onto the escape direction (pressure, scattering)
$w = 1 - e^{-\sqrt{(\tilde{\theta}\tau_D)}}$	weight: 1 co-breathing, 0 bound

F is the vector field of the motion, I the identity. The **mode** ξ is an oscillation form of the whole whose pressure direction is closed. The **weight** w says how strongly it acts, and it depends on the expansion rate $\tilde{\theta}$ of the region: if the region breathes along, $w \approx 1$. If it is bound, w goes to zero. As long as a region breathes along ($w \approx 1$), the mode is read like pressureless dust with the weight

$$Q = q/(1 - q) = \Omega_c/\Omega_b$$

By Kac, $1 - q = \Omega_b/\Omega_m$ is the measure of the open region and $1 + Q$ the mean return time of the whole. What cosmology reads as dark matter Ω_c is this mode. Linearly this is identical with the standard model. The mode has no initial data of its own, it starts in the state of the baryons. From the turnaround point of a region onward, it passes into the local rule, about one radian of the orbit after the turnaround. The balance of whole and place closes at the **seam** $R_Q = r^*/(-\ln q)$, where the local q reaches the cosmic one ($a_N = 0.029\cdot a_0$, Milky Way $R_Q = 52.0$ kpc). There $\chi = Q\cdot(-\ln q) = 0.917$, almost the deep limit, inside the local rule acts, outside $1 + Q$ counts as the measure of the whole. In SPARC 131 points lie beyond the seam, their residual against the local rule is $+0.013 \pm 0.006$ dex, capped at $1 + Q$ they would lie 0.094 dex above. Likewise the lensing around isolated galaxies follows the local rule up to 2.6 Mpc. In clusters the mode is not handed over: their acceleration follows the measure of the whole, $(1 + Q)\cdot g_{bar}$, down into the core, not the local rule. It is the same formula with the return number at the nome of the whole, $n_B(q) = q/(1 - q) = Q$, instead of at the nome of the place, and at the seam both are equal. The mode passes over only where the phase-space sheet folds, and gas folds only once it has given off its heat: cold gas breaks up into clouds whose orbits cross, hot plasma holds itself up by its pressure, and the sheet does not fold. The plasma of clusters has never given off its heat, in the core its cooling takes 13.6 to 38 Gyr (X-COP). Because the folding cannot be undone, the return of extinct systems keeps reading what they once bound. In HIFLUGCS (10 clusters, masses from the motion of the cluster galaxies, Li et al. 2023) $(1 + Q)\cdot g_{bar}$ lies +0.03 dex on the data over all 108 points ($\chi^2 = 601$), the local rule 0.17 dex below ($\chi^2 = 4173$). In X-COP the local rule demands in the core 12 times the gas in baryons, the measure $1 + Q$ only 2.9 times. Co-breathing, the mode feels no pressure: it follows the motion of pressureless dust and mixes its orbits during infall, while the gas keeps its energy as heat. That is why it lies more concentrated than the gas. In the shape after infall (NFW) it carries the core without the stars of the central galaxy: X-COP reads the concentration $c = 4.2$, HIFLUGCS $c = 5.5$ ($\chi^2 = 447$ instead of 601). Where the gas never cooled, the mode computes like cold dark matter, and the observer reads the halo of the standard model. At recombination, linear perturbations lie above the threshold but breathe along ($w \approx 1$) and are read like dust with $1 + Q$. The local rule sets in for each region only at its turnaround.

Q per radian. The measured value is $Q = 5.36 \pm 0.06$. The model counts per radian, as in $\hbar$ and $\lambda\!\!\!^{-}$: the whole is one full turn, the open part, the baryons, one radian of it (the author's counting, the derivation is still outstanding, Section 31):

$$\nu_{cosm} = 2\pi \qquad \Omega_b/\Omega_m = 1/2\pi \qquad Q = 2\pi - 1 = 5.283$$

This lies 1.3 σ from the measurement. The geometry of the whole (sound angle, sound wave and distances with free a_0 and ω_b) reads $Q = 5.22 \pm 0.05$, with the height of the peaks as a fourth reading 5.26 ± 0.04: $2\pi - 1$ lies $\Delta\chi^2 = 0.2$ away, the measured ratio 5.0. With all precision data the standard model itself reads $Q = 5.278$. In the model itself, with H_0 from a_0 and the maximum of the dark energy at the saturation of the loop, all precision data read $Q = 5.36 \pm 0.05$ and $a_0 = 1.076 \pm 0.006$: $2\pi - 1$ lies $\Delta\chi^2 = 2.4$ away. Independently the clusters read the same measure: with the mode in the shape after infall HIFLUGCS reads $Q = 5.5 \pm 0.3$. From below, the form closes: the return depth of all baryons of the cell must not reach beyond Λ, $r^*/\Lambda = \sqrt{(6/(1 + Q))} \leq 1$, hence $Q \geq 5$. The measured value is $r^*/\Lambda = 0.971$, so the whole lies just below its own meeting point. The same counting per radian closes the atom: the wave of the electron closes after one turn, $2\pi \cdot a_B = h/(m_e \cdot v)$. The roles are mirrored: for the whole, closure sets the ratio (Q) and the content sets the length (Λ). For the atom, closure sets the length (a_B) and the content, the charge, sets the ratio α. The coupling α is content, not form, like the amount of baryons.

25 The Breathing and the Dark Energy

The growth of distances is the **breathing** of the medium: the loops stretch, $L(t) = \bar{L} \cdot a(t)$, with a(t) the measure of the stretching and $\bar{L}$ the length it refers to. The wavelength of light is a loop length and breathes along, $1 + z = L(t_{ob})/L(t_{em})$. The loops breathe alike: beyond $z = 0.023$, Pantheon+ shows no preferred direction, the dipole is $\Delta H/H = 0.9$ % (0.7 σ). A quadrupole appears only at $z > 0.1$, where the supernovae come from a few fields of the sky, and falls to 1.8 σ with a zero point of its own per sub-catalogue.

The vacuum part, derived. The action of the field carries the same function as the rule, on the third step:

$$\tilde{Q}(s) = -4 \int_s^\infty x^3 \cdot n_B(x)\, dx \qquad \tilde{Q}(0) = -4\pi^4/15$$

The constant $-4\pi^4/15$ depends neither on single funnels nor on the smoothing: it is the restoring of the closed whole. In the action it appears with the measure of the inner direction, $1/\ell_0^2$ with $\ell_0 = 2\Lambda$, doubled for the outward and return path of the return and as a restoring with reversed sign: $(2/\ell_0^2) \cdot 4\pi^4/15$, twice a constant per closure length, $\Lambda_s = \pi^4/(15\Lambda^2)$. Read as an energy density, with $\Lambda = c^2/(2a_0)$:

$$\rho_s = \Lambda_s \cdot c^2/(8\pi G) = \pi^3 c^2/(120 \cdot G \cdot \Lambda^2) = \pi^3 a_0^2/(30 G c^2)$$
$$\Omega_s = \rho_s/\rho_{crit} = (\pi^2/45) \cdot \Theta^2 \qquad \Theta = 2\pi a_0/(cH_0) = 1.020 \qquad \Omega_s = 0.228$$

This is a vacuum part with no number other than a_0, with the equation of state $p = -\rho c^2$ ($w_{DE} = -1$, this w_{DE} is not the weight w of the mode). Because Λ has been fixed since $z \approx 10$, it is constant. Early on, when radiation contributed more to the fold, a_0 was smaller and the vacuum part is absent, as the background radiation requires. If it alone carried the expansion, this would run into the final state $H_s = H_0 \cdot \sqrt{\Omega_s} = 32.1$ km/s/Mpc, with three parts into 55.6 km/s/Mpc. Because the fold is fixed, Λ, a_0 and ρ_s remain fixed while the loops stretch. Θ runs from 0 to the final value 1.23 and stands at 1.020 today. Today's coincidences, such as $a_0 \approx cH_0/2\pi$, are the present inflation state of the balloon, not numbers to be hit. It is itself a mirror: its energy density is the geometric mean of the Planck density and the quantum density of the closure length, $\rho_s \cdot c^2 = (\pi^3/30) \cdot \sqrt{(c^7/(\hbar G^2) \cdot \hbar c/\ell_0^4)}$. ħ drops out. In Planck units it therefore lies at ε_L^2, $\Lambda_s \cdot \ell_P^2 = (4\pi^4/15) \cdot \varepsilon_L^2 = 0.94 \cdot 10^{-122}$, exactly midway between one and ε_L^4. The often-quoted vacuum energy that is "10^{122} times too small" is this midpoint.

Three loops. The whole too splits into the triad: the centre is the flux out of the sources, the fold of the baryons. The vortex is the return at the place, the dark matter. The background is the winding classes of the $b_1 = 3$ loops of the torus, the restoring: the **dark energy**. Each loop restores the share of the fold in its direction, because $\Delta s = GM/(3c^2)$ is the share of one of the three spatial directions, and together this is the whole fold, $3 \cdot \Sigma \Delta s_b = \Sigma GM_b/c^2$, nothing twice, nothing missing:

$$\rho_{DE} = b_1 \cdot \rho_s = 3 \cdot \pi^3 a_0^2/(30Gc^2) \qquad \Omega_{DE} = b_1 \cdot (\pi^2/45) \cdot \Theta^2$$

What counts is the length of each loop (magnitude), not its winding (phase). The weight is a quantity of the inner direction and holds for every shape of the cell. The 3 of the fold per direction (d = 3, at the place) and the 3 of the loops ($b_1 = 3$, in the whole) are two roles of the same dimension. Confusing them would be an observer's fallacy.

One threshold for galaxies and cosmos. Like dark matter, dark energy thus depends only on a_0. From ω_b, ω_c, T_{CMB} and a_0 follows the breathing: with $a_0 = 1.06 \cdot 10^{-10}$ m/s^2: $H_0 = 67.2$ km/s/Mpc and $\Omega_{DE} = 0.685$, with $1.1 \cdot 10^{-10}$: $H_0 = 68.9$ and 0.700. Conversely, reading a_0 from the measured distances alone (Pantheon+, 1590 supernovae, DESI DR2 BAO, with ω_b and ω_c from Planck) gives $a_0 = 1.09 \cdot 10^{-10}$ m/s^2 (1.07 to 1.11), and with the model's Q 1.08 (1.06 to 1.10). Rotation curves (1.11), sound wave (1.10) and distances (1.09) read the same threshold: the threshold of the single funnel is that of the whole. The breathing, read after Planck, lies below at 1.065. That is at the same time the upper limit for a positive state part and lies within the band of the galaxies. Like Q, only a ratio without a clock is fixed in time: $k_{DE} = \rho_{DE} \cdot Gc^2/a_0^2$, the vacuum part has $k_s = \pi^3/30 = 1.034$. Quantities with H_0 or the age of the universe hold only in one epoch. Today $\Omega_{DE}/\Omega_s = 3.03$ (read after Planck).

In the distances. The form part, the restoring alone with $w_{DE} = -1$ (the state part, which depends on the state of binding, follows in Section 26), with no free number ($\Omega_m = 0.315$), lies only $\Delta\chi^2 = 2.0$ above the fitted standard model ($\Omega_m = 0.304$), 1.4 σ. A dark energy that weakens late (equation of state $w_{DE} = w_0 + w_a \cdot z/(1 + z)$ with $w_0 = -0.89$, $w_a = -0.21$) lies $\Delta\chi^2 = 6.9$ below it, 1.8 σ: the direction of the state part, weak in these data alone. With $H_0 = 67.2$, DESI requires the sound horizon $r_d = 149.0$ Mpc. Without any calibration, because r_d and H_0 cancel, DESI reads the shape of the expansion in the ratio of transverse to radial distance, D_M/D_H, at six redshifts (z = 0.51 to 2.33). The form part meets it with $\chi^2 = 6.8$. The absolute scale has no free number either: early on the mode acts linearly like dust, the **sound horizon**, the distance the sound wave travels in the early plasma until it ends, follows from ω_b and ω_c alone ($r_d = 147.1$ Mpc), and H_0 and Ω_m follow from a_0. DESI then reads $a_0 = 1.10 \cdot 10^{-10}$ m/s^2 (1.09 to 1.11), with $H_0 = 68.9$ at $\chi^2 = 10.4$ for 13 values. The standard value 1.06 lies $\Delta\chi^2 = 27$ away. With the model's Q, $2\pi - 1$, instead of the measured one, DESI likewise reads $a_0 = 1.10$ ($r_d = 147.6$ Mpc, $H_0 = 68.9$, $\chi^2 = 10.3$): then the whole scale follows from ω_b, T_{CMB} and a_0. Reading Q itself from the distances (DESI with fixed scale, Pantheon+ in shape, a_0 free) gives Q = 5.5 (5.3 to 5.7). $2\pi - 1$ lies $\Delta\chi^2 = 0.9$ away: without the measured ω_c, with ω_b alone, the distances carry the same weight of the mode.

At the background radiation. Early on the mode acts linearly like dust. The spectra of the background radiation (Planck), its lensing, DESI, Pantheon+, the growth and the light nuclei are therefore computed by the standard calculation with $\omega_c = (2\pi - 1) \cdot \omega_b$ and H_0 from a_0, with one free number fewer than the standard model. On the spectra alone the model lies $\Delta\chi^2 = 1.0$ above the standard model, on all data 0.01. With the state part (w_0 and w_a free) it gains $\Delta\chi^2 = 9.6$: $w_0 = -0.84 \pm 0.05$, $w_a = -0.56 \pm 0.18$, the maximum of the dark energy lies at $z = 0.40 \pm 0.07$, and the threshold reads 1.070 ± 0.013, the standard value only $\Delta\chi^2 = 0.8$ away. With $a_0 = 1.06$ and the state part the model has as many free numbers as the standard model and lies

$\Delta\chi^2 = 8.8$ below it. If the maximum lies where the loop saturates, where three parts catch up with the fold (w_0 then follows from w_a, one number fewer), the data meet this just as well ($\Delta\chi^2 = 0.9$) and read the dark energy today as $a_0 = 1.085 \pm 0.003$, like the geometry of the whole. The maximum then lies at $z = 0.32$. What is read is form and state together: as the whole dark energy today the standard value 1.06 does not work ($\Delta\chi^2 = 48$), as the form part with the saturation at it the data meet it just as well, and the state part today is 5.4 ± 0.7 % of the form part. So the model with free a_0 has as many free numbers as the standard model and lies $\Delta\chi^2 = 8.7$ below it.

H_0 of the ladder. Calibrated with Cepheids, Pantheon+ instead reads $H_0 = 73.5 \pm 1.0$, in every interval up to $z = 1$ between 73.0 and 74.5. The difference from the breathing is therefore not a local state of the stretching. It lies in the calibration of the ladder, which places the supernovae 0.20 mag brighter, against the scale of the sound wave ($H_0 = 68.9$) 0.14 mag.

26 The State Part

Besides the fixed form part there is a part that depends on the state: the **escape measure of binding**. Binding makes escape directions of the state space inactive. The inactive directions pass into the return and act as a stretching of the loops. The same measure acts at once as binding in the region and as stretching in the whole, two sides of the same fold. The state part is carried by the baryons with orbits of their own, the bound gas whose mode has been handed over, not the stars (a star is a single funnel) and not the plasma of the clusters, in which the mode has not been handed over. Both parts hang on the baryons, on different sides: the vacuum part on their fold, with $\Lambda = (1 + Q)\cdot\Sigma\Delta s_b$ one has $\rho_s \propto 1/((1 + Q)\cdot\Sigma\Delta s_b)^2$, more fold means a longer return and a weaker restoring, whether gas or star. The state part on their binding. In SPARC ($Y_d = 0.67$), below $v_{flat} = 40$ km/s 72 % of the baryons are gas, above 150 km/s 12 % (median): the small, gas-rich regions carry the state part. What is counted is the accessible state space, not the place: per baryon the largest region counts, a sub-binding within it adds no measure. Hence only regions that are themselves the largest can lose, single galaxies, not galaxies in groups.

Size and lower limit. In the deep regime the inactive measure of a region is $(R_{ta}/r^*)^{3/2}$, with the turnaround radius $R_{ta} \propto M^{1/3}$ and $r^* \propto M^{1/2}$, per baryon $\propto M^{-1/4}$: small and late-bound regions carry more. The lower limit is set by the gas itself: only what thermal motion holds becomes bound, in the deep regime $M_{Gas} = (81/4)\cdot\sigma^4/(Ga_0)$, independent of density. With the gas pressure of the slow dwarfs (SPARC, 8 to 10 km/s), $M_{Gas} = 0.6$ to $1.4\cdot10^7$ M⊙. Written without mass, a region holds only gas with $\sigma < (4/81)^{1/4}\cdot v_{flat} = 0.47\cdot v_{flat}$. With 8 to 10 km/s there is no gas-bearing disc below $v_{flat} = 17$ to 21 km/s. In SPARC the smallest measured rotation is 17.8 km/s, the smallest flat one 18.6 km/s.

The whole is thus not an equilibrium but a **feedback loop**:

fold (fixed) → restoring → stretching of the loops
stretching ⊣ binding (stretching keeps regions open)
binding → inactive escape directions → return → stretching

More stretching keeps regions open and so lowers the state part: the loop limits itself. Its lag is the time until the turnaround of a region, a time in space, not τ_D. Q and the vacuum part are form (nome of the cell, Λ), fixed in time. The escape measure is state and follows the bound measure of the largest region in each case. An amount "today" is a phase of the loop, not a property of space. The measurement reads form and state as one fixed dark energy: Ω_{DE} presupposes a fixed dark energy and $\Sigma\Omega = 1$, w_0 and w_a a smooth fluid, Ω_c reads

the mode ξ as dust, and along the light cone the observer sees earlier phases of the same loop. The loop saturates where dark energy catches up with matter. The vacuum part alone would catch up with it only at $z = -0.10$, three parts do so at $z = 0.30$. The small state part hardly shifts this, the moment hangs on a_0 alone. After that, the backflow dominates: gas is heated above the binding of its region and releases its inactive measure, regions merge, gas turns into stars. Dark energy has a maximum and weakens afterwards. This is the direction the distances weakly show.

Two paths of backflow. *Merging:* when a region falls into a larger one, its escape directions are already bound in the parent field and cease to be escape directions. The number of turnarounds remains. Whether this raises or lowers the state part depends on the slope n of the spectrum: typically, regions $M_z \propto (1 + z)^{-6/(n+3)}$ bind at time z, so $R_{ta}/r^* \propto (1 + z)^{-(n+2)/(n+3)}$. Below $n = -2$ merging lowers the state part, above it raises it. The boundary lies at 3 to $4 \cdot 10^{12}$ M⊙, set by the equality of radiation and fold at $z_{eq} = 3402$. *Heat:* when gas is heated above the binding of its region, it leaves the seam and its inactive measure is released. Gas at 100 to 300 km/s is held only by regions from $1.4 \cdot 10^{11}$ to $1.2 \cdot 10^{13}$ M⊙. If gas ΔN leaves the seam, the inactive measure drops by $\Delta N \cdot (\ln B - \frac{3}{4})$, with B the coefficient of the region (the author's derivation). When the vacuum part halts the growth of structure, new binding ends but heating does not: then the backflow dominates.

27 Cycle, Memory and Entropy

On the largest scale the **cycle** closes: at the bottleneck the medium condenses into fold again. The total entropy grows in every round, and the cosmos runs a spiral, not into a heat death. The cycle closes in form, not in state.

Memory. Folded space does not forget: every vortex leaves a lasting crease, and new systems use the existing grooves. The crease is form, not a store beside space. In the cosmos it is a state of the closure rule: what lasts is its weight w, which depends on the smallest expansion rate in the history. A region once bound stays bound. Memory is the number of turnarounds of a region: its phase runs only forward under a restoring slope, the weight drops as a step at the first turnaround. What is protected is the count, not the single loop.

Entropy and heat. The bottleneck is, at the place, the fold $r \le 6\Delta s$ and, in the whole, the horizon. On both levels the same form of entropy holds, $S = 36\pi k \cdot (\Delta s/\ell_P)^2 = 4\pi k \cdot (M/m_{Pl})^2$, for the Sun $1.05 \cdot 10^{77}$ k. The temperature of the fold is $kT = \hbar c/(24\pi\Delta s)$, that of the horizon twice as high for the same fold. Heat is here a loop length: the thermal length ħc/kT of every closure is its loop, for the fold $24\pi\Delta s = 4\pi \cdot r_0$, for the horizon $2\pi c/H$, for the inner direction $\ell_0 = 2\Lambda$ ($kT_I = \hbar a_0/c$). The entropy is the coupling, $S = 4\pi k \cdot (M/m_{Pl})^2$. For the horizon $T = \hbar c/(12\pi \cdot k \cdot \Delta s_H)$ with $\Delta s_H = c/(6H)$. Order grows inside only as long as more entropy is exported than produced. This is syntropy, $Syn = S_{eq} - S$, with S_{eq} the entropy in equilibrium. It changes as $dSyn/dt = J - \dot{S}_i$, with the entropy export J and the production $\dot{S}_i \ge 0$. At the bottleneck of the whole the medium condenses into fold again. Λ grows and the vacuum part falls. The entropy belonging to the vacuum part depends, like that of a fold, only on the square of a fold length, $S_\Lambda = (45/\pi^3) \cdot k \cdot (\Lambda/\ell_P)^2$, and requires $\dot{\Lambda}/\Lambda \ge \dot{S}_i/(2S_\Lambda)$. This minimum rate is tiny and does not act on the dark energy today. For gravity, compression increases entropy: the fold is, in the small, the place of greatest entropy and, in the large, of the simplest form.

28 The Web

The torus has Euler characteristic zero. Therefore every form of the slope obeys

$$\text{funnels} - \text{filament saddles} + \text{wall saddles} - \text{voids} = 0$$

The closed medium enforces the web of nodes, filaments, walls and voids. Between two funnels there is always a filament saddle, between two voids a wall saddle. Fewer than 1, 3, 3, 1 is impossible, the Betti numbers of the torus: the simplest form, cos x + cos y + cos z, has exactly one funnel, three filaments, three walls and one void. The three filaments of the smallest web are the three loops that carry the dark energy. Reversing the slope exchanges funnel with void and filament with wall: the web is a mirror that only the fold breaks, because only the funnels carry fold.

Part VI: Reading and Testing

29 The Observer's Fallacy

When the rule of the whole changes, a measured value at a place deviates, and the observer invents a local thing as its carrier. In the model nothing behaves differently. The rule of the medium has changed. For every quantity one must therefore ask in which direction of view it holds: **at the place** (L, in space), **in the inner direction** (I, orthogonal to space: return, Λ, ϖ) or **for the whole** (G, the torus as a cell). And in which state: breathing along or bound, in which epoch, for which observer.

Phenomenon	read as a thing	as a state of the medium
mass	substance in space	fold depth Δs
gravity	emitted force	slope g
time	a flow of its own	difference in unrolling δ
expansion	bodies fly apart	the loops stretch
dark matter	invisible particle	changed rule of motion below the threshold a_0
dark matter in the cosmos	dust with its own density Ω_c	mode ξ of the closure rule with weight Q
dark energy	a form of energy of its own	restoring of the three loops and escape measure of binding
vacuum energy	energy of empty space, 10^{122} times smaller than expected	mirror between Planck density and closure length
temperature of the return	heat	period of the return in the inner direction
entanglement	signal through space	shared node in the inner direction
black hole	hole with a singularity	fold: shell at the centre, finite fold Δs

Rule instead of source. The difference between an additional substance and a changed rule can be stated precisely. Closure is an **operator** $\mathscr{P}$ acting on the generator of motion, $L \to \mathscr{P}L$, with $\mathscr{P}_q = (1 - q)\cdot\Sigma\, q^m T^m$, T the step by one return (one turn in the inner direction) and q^m its weight after m returns. It deletes no degrees

of freedom and preserves the measure, while certain directions become inactive and pass into the return. In closed form $\mathscr{P}_q = (1 - q)(I - qT)^{-1}$: for $q \to 0$, $\mathscr{P} = I$, open and Newtonian. For $q \to 1$, space is closed. On the torus, applied to a power kernel, $\mathscr{P}$ is the lattice sum after Eisenstein. Its lowest series over the return steps of one direction is $\Sigma q^n = q/(1 - q)$, the addition of the circular orbit. The bound measure grows only as long as a parameter wanders. A source, by contrast, adds density:

	Source (dark matter as substance)	**Operator $\mathscr{P}$**
where it acts	$\rho \to \rho + \rho_{DM}$	$L \to \mathscr{P}L$
potential	additive	multiplicative
localisable	yes	no
degrees of freedom	additional	existing ones become bound
time structure	without memory	memory (weight w, handover of the mode)

The density read off, $\rho_X = \nabla\cdot(a - a_N)/(4\pi G)$, is an output of the calculation and not even unique: different distributions produce the same external field. If all tests run alike, source and operator are just two descriptions. If one test runs robustly apart, they can be told apart.

Entanglement. Two entangled particles may lie arbitrarily far apart in space. In the inner direction they share the same node. No signal travels. Both measurements act on the same node. Their correlation is the fundamental vibration of the winding, $E(\Delta) = -\cos\Delta$ with Δ the angle between the two measurement directions, and gives $S = 2\sqrt{2}$ (S here Bell's measure of correlation, not the entropy). As a property given along, $S \le 2$ would remain.

30 What Can Be Tested

- Wide binary stars remain Newtonian: stars are single funnels. At the place of the Sun a star of 1 M☉ has the smoothing length $\ell = 349$ kAU, wider than the Jacobi radius of a pair of 1 M☉ (277 kAU). In general the heavier star of every pair reaches with $\ell^3 = Gm/\mathscr{T}$ at least to the Jacobi radius of the pair, $r_J^3 = G(m + m')/(2\mathscr{T})$ for a flat curve.
- In disc galaxies the dynamical mass inside is larger than the lensing mass. For spheres and outside both are equal.
- The threshold does not wander: early galaxies (z = 1 to 2) have the same a_0.
- Galaxies in clusters follow the same relation inside as free ones. Only the tide disturbs them.
- When two clusters collide, pressure brakes the gas but not the mode: it runs on with the galaxies, and the lensing separates from the gas. At the galaxies it reads at least Q times the collided gas, at the gas only the gas's own fold.
- Counter-rotating stellar discs of the same galaxy follow the same relation: the one sense of circulation of the return is a sense of time, not of rotation.
- The largest additional acceleration lies at $a_N = 2.54\cdot a_0$ and amounts to $0.648\cdot a_0$.
- Dark energy weakens late ($w_0 > -1$, $w_a < 0$).
- The next closure number of nuclei is 184.
- Hot gas in equilibrium shows the same enhancement as the stars of the same galaxy.

- At the Sun and Earth saddle point there is no additional acceleration: both bodies remain Newtonian up to their smoothing length.
- The growth of structure departs from the standard model only below the turnaround scale, the size of the regions that detach from the expansion today (Section 26).
- The vortex of the rule can be tested in discs by $\partial_z a_R \neq \partial_R a_z$.
- The form of the return can be rebuilt in the laboratory: an RC chain whose tap n_R is fed back to the input is the renewal $u(0) = a_N + u(\varpi_R)$ and gives the Bose form exactly, $\nu = 1/(1 - e^{-s})$ with $s = n_R \cdot \operatorname{arcosh}(1 + RG/2)$. At $R \cdot G = 0.01$ (R the resistance, G the conductance per link) and $n_R = 20$, $\nu = 1.1567$. This tests the equation, not gravity: in the laboratory $s \approx 3 \cdot 10^5$, there is no laboratory prediction that only the model makes.

31 What Is Open

- **The reading of Λ** as the entire fold of the cell is posited, supported by the mirror between part and whole.
- **Q** is, counted per radian, $2\pi - 1$ (the author's counting, the derivation is still outstanding), 1.3 σ from the measurement. Whether counting per radian is compelling remains open.
- **α and m_p/m_e** are content, not form. They follow from no equation of the model.
- **The state part:** the data read its course, open is its size, the share of bound gas.
- **H_0 of the Cepheid ladder** lies above the value of the breathing at every z. Whether its calibration is to be read differently in the model will only be shown by the Cepheid data themselves.
- **The warp of the Milky Way:** does not fit yet.
- **Shear:** in KiDS-1000 the model and ΛCDM lie equal, both demand a lower amplitude. Whether the mode handed over in galaxies lowers the growth below the turnaround scale is open.
- **Ultra-diffuse galaxies:** six gas-rich ones rotate like Newton alone, the rule demands two and a half times the velocity. Their outer point lies at the seam, where the local rule and the measure of the whole give the same, so a mode that was not handed over does not carry them either. Gas-rich isolated dwarfs at the same place follow the rule, their dispersion needs no inclination. Their inclination is small and read only from the shape of the gas.
- **Red lenses:** around isolated galaxies that no longer form stars the lensing lies above the rule of their stars. Whether the gas of their past carries it as for the dwarfs, only their metallicity will show.
- **Elements beyond 118:** the tipping point of the nucleus in the valley of stability (the neutron number per proton number with the largest binding) is computed, not measured (Section 18). Open are the filling of the g shell and the lifetime against fission.

32 Conclusion

The treatise began with a funnel: a drawing in which more space within space appears as depth. From it followed slope, clock rate and energy. From the closedness of space, the return, the threshold and the rule. From the shape of the torus, the three generators mirror, threefold and full turn. The same forms recur on every level, from the Planck point through nucleus, atom, star and galaxy to the whole, tested on nuclei, atoms and substances. Galaxies, sound wave and distances read the same threshold. The whole sets the scale of the parts, and the parts fold the whole.

Quod est inferius est sicut quod est superius, et quod est superius est sicut quod est inferius, ad perpetranda miracula rei unius. (Tabula Smaragdina)

Appendix A: Symbols

Symbol	Name	Meaning
$\Delta s = GM/(3c^2)$	fold	extra length in the centre, measure of mass
$g = 3c^2\Delta s/r^2$	slope	steepness of the funnel wall, the weight we feel, $g_N = a_N$ the Newtonian one, in e·g the magnetic charge, g = 2 the ratio of magnetic moment to angular momentum
$d\tau/dt = \sqrt{(1 - 6\Delta s/r)}$	clock rate	unrolling at the place against flat outer space, τ the proper time at the place
δ	spread	distance of the clock rate from one, on orbits v^2/c^2, in $1 - q \propto \delta^{1/2}$ the distance to the tipping point
$r_0 = 6\Delta s$	limit radius	where the clock stands still
fold	black hole	funnel whose shell lies at the centre, $R = r_0$
$\bar{\lambda} = \hbar/(Mc)$	quantum length	mirror partner of the fold
ℓ_P, m_{Pl}	Planck length, Planck mass	centre of the mirror, meeting point of fold and quantum length
ϖ	inner direction	its coordinate, the smoothing depth of the return
$\varpi_R = r^*$	return depth	depth at which the state returns, equal to the threshold radius
$\Gamma = a_N/c$, θ	losses	from the clock-rate gradient and from the expansion of the region, $\Gamma(k)$ the gamma function
$\tilde{\theta}$	expansion rate	of the region, the loss θ of the return equation
$D = GM/c$	spreading	in the inner direction
Λ	closure length	where the inner direction closes
$\tau_D = 2\Lambda/c$	return time	mean return time, the same for every mass
$a_0 = c^2/(2\Lambda)$	threshold	slope below which the return acts
$r^* = \sqrt{(GM/a_0)}$	threshold radius	transition to the background
$q = e^{-s}$, $s = \sqrt{(a_N/a_0)}$	return fraction	share that comes back
$\nu = 1/(1 - q)$	enhancement	sum of all returns
$\chi = s/(e^s - 1)$	switch	from Newton (0) to the deep regime (1)
ξ, w	mode, weight	closure rule of the whole
$Q = \Omega_c/\Omega_b$	weight of the mode	mean return time of the whole minus one
$b_1 = 3$	number of loops	independent loops of the torus
ρ_s, Ω_s	vacuum part	restoring per loop
λ_H	holding length	distance at which a held funnel holds its building blocks
τ	modulus	shape of the torus, in $d\tau/dt$ the proper time
$n_B = 1/(e^s - 1)$	Bose occupation	number of stimulated returns
$\tilde{m} = m_{Pl}^2/(2M)$	mirror mass	mass whose quantum motion is the return
$kT_I = \hbar a_0/c$	period of the return	period of the inner direction, written as a temperature, not heat at the place

Symbol	Name	Meaning
$\varepsilon_L = \ell_P/\ell_0$	ladder number	fixes the rungs of the ladder of lengths
$\Theta = 2\pi a_0/(cH_0)$	breathing number	threshold against today's expansion rate
d_{eff}	flux dimension	dimension the rule leaves to a form
k	coupling	strength with which two cycles lock in, in kT the Boltzmann constant
φ	phase	of the coupling, k·cos φ its lossy part
$j = a_J/(3\Delta s)$, $a_J = J/(Mc)$	spin number, twist	rotation of a funnel with angular momentum J, j = 1: both horizons coincide
counting rule	closure per sense of circulation	every state holds one winding per sense of circulation, it holds white dwarfs and neutron stars
l, κ, Ω, ν_z	orbital winding, frequencies	number of nodes of an orbit, frequencies of the radial, the orbital and the perpendicular oscillation
$\ell_0 = 2\Lambda$	length of action	measure of the inner direction
M_{min}, M_c, M_U	rungs of the ladder	$\varepsilon_L \cdot m_{Pl}$, $\varepsilon_L^{1/3} \cdot m_{Pl}$, m_{Pl}/ε_L
$M_* = m_{Pl}^3/m_p^2$	mass scale of stars	1.853 M⊙
$M_{Gas} = (81/4)\cdot\sigma^4/(Ga_0)$	smallest gas-bearing mass	what the thermal motion σ of the gas holds
R_{ta}, R_Q	turnaround radius, seam	where a region detaches from the expansion, where the local q reaches the cosmic one
L	cell	spatial cell of the torus, 27.9 Gpc
$\Lambda_s = \pi^4/(15\Lambda^2)$	constant per closure length	the vacuum part in the action, $\rho_s = \Lambda_s \cdot c^2/(8\pi G)$ its density
$Q_{el} = n \cdot e$	charge	winding n of the phase of a node
Ω_b, Ω_c, Ω_m, Ω_{DE}, ω_b, ω_c	densities	shares of the baryons, the mode, matter ($\Omega_m = \Omega_b + \Omega_c$) and dark energy in the critical density ρ_{crit}, $\omega = \Omega \cdot h^2$ with $h = H_0/(100$ km/s/Mpc)
Y_d	mass-to-light	conversion of light into fold, 0.67
$\mathscr{T}$	tide	of the smoothed parent field
T	step, temperature, period	step by one return in the operator $\mathscr{P}$, in kT the temperature (T_U after Unruh, T_I of the return, T_{CMB} of the background radiation), in $T^2 = 4\pi^2 r^3/(3c^2\Delta s)$ the orbital period, T^n the torus of n circulations
B	coefficient, binding	coefficient of the backflow, ln B, in $Z \cdot E_H + N \cdot E_n - B$ the binding of the nucleus
$x = E_C/(2E_S)$	fissility	winding against holding in the nucleus
$\Sigma_0 = a_0/(\pi G)$	threshold of surface density	242 M⊙/pc^2, the centre of a disc at the threshold
N(N + 1), v	ladder of periods	couplings of the mirrored field of the shell, N = n + l, and its height $3.44 \cdot Z^{2/3}$
L_{Edd}	Eddington limit	radiative transport in the star, $(9/2) \cdot m_p \cdot c^3 \cdot \Delta s/r_e^2$

Symbol	Name	Meaning
$k_{DE} = \rho_{DE} \cdot Gc^2/a_0^2$	time-fixed number	ratio without a clock, vacuum part $k_s = \pi^3/30$
c_{eff}	light from outside	$c \cdot \sqrt{(1 - 6\Delta s/r)}$, delay of the unrolling
$\mathscr{P}$	closure operator	acts on the generator of motion, not as a source
G, c, ħ, e	constants of nature	gravitational constant, speed of light, quantum of action $\hbar = h/2\pi$, elementary charge
M, m, r, R, t	mass, distance, time	mass of the body (m that of the particle, $M\odot$ the solar mass), distance from the centre, envelope, time
m_p, m_e, m_u	masses	proton, electron, atomic mass unit
Z, N, A	nuclear numbers	protons, neutrons, mass number A = Z + N, N also the number of building blocks of a held funnel and N = n + l the step of the shell
n	number of the shell	n + l the step, in $Q_{el} = n \cdot e$ the winding, in Σq^n the number of returns
H, H_0	expansion rate	today H_0 (km/s/Mpc), $\Delta s_H = c/(6H)$ the fold of the horizon
z	redshift	$1 + z = L(t_{ob})/L(t_{em})$, in ż the state, in $u = z^{(a+3)/2}$ the place in the plane
a(t), $\bar{L}$	stretching	measure of the stretching of the loops, $L(t) = \bar{L} \cdot a(t)$
σ	balance, thermal motion	$\Sigma\Delta s - \Delta L = 0$, σ, σ_v the speed of the thermal motion of the gas, as a measure of measurement the standard deviation
S	entropy	S_{eq} in equilibrium, S_Λ of the vacuum part, in V/(S·R) the surface, in $S = 2\sqrt{2}$ Bell's measure of correlation
E	energy	E_C, E_S Coulomb and surface energy of the nucleus, E_H, E_n rest energies of hydrogen atom and neutron, E_G the Gamow energy, E(Δ) the correlation
u, G(ϖ)	return	share of the state at depth ϖ, G(ϖ) its stationary profile
Φ, Φ_N, A, Ψ	potentials	potential of the metric, Newtonian potential, potential of the vortex, second potential of the metric
F, I	generator, identity	vector field of the motion (F_{par} the parametric part), identity operator
b, p	ray, orbit	distance of the light ray from the centre, parameter of the orbit, p also the number of the period
R_d, M_b	disc	disc scale length, baryonic mass of a galaxy
μ, m̄	moments and masses	magnetic moment (μ_B the Bohr magneton), line mass of a filament, mean mass per particle (μ_e per electron) in m_u, m̄ the mean particle mass
ζ, P, O(k)	functions	Riemann's zeta function, associated Legendre function, states of the held body up to shell k
T_α, Q_α	alpha decay	half-life and energy of the decay
ψ, Δω, W	cycles	phase difference of two cycles, difference of their eigenfrequencies, frequency band
i, ln, lg	signs of calculation	imaginary unit, natural and decimal logarithm

Symbol	Name	Meaning
AU, pc, fm, dex, mag	units	Earth and Sun distance, parsec (kpc, Mpc, Gpc), 10^{-15} m, power of ten of the logarithm, magnitude

Appendix B: Data

All numbers in this treatise are computed in the model and checked against the simulation. The measurements come from five data sets: SPARC (rotation curves of 175 galaxies, the computation uses the 141 whose curves have quality 1 or 2 and whose inclination is at least 30°), AME2020 (nuclear masses), PubChem (periodic table), Pantheon+ (supernova distances) and DESI DR2 (distances from the sound wave of the early cosmos). The cosmic densities ω_b, ω_c and the temperature of the background radiation are taken from measurement (Planck 2018).

Appendix C: Sources

Data.

- SPARC: F. Lelli, S. S. McGaugh, J. M. Schombert (2016), rotation curves and photometry of 175 disc galaxies.
- AME2020: Atomic Mass Evaluation 2020, nuclear masses and binding energies.
- PubChem: periodic table of the elements (ionisation energy, atomic radius, melting and boiling points).
- Pantheon+SH0ES: D. Scolnic et al. (2022), D. Brout et al. (2022), distances of type Ia supernovae with covariance.
- DESI DR2: DESI Collaboration (2025), distances from baryon acoustic oscillations.
- Planck 2018: cosmological parameters ω_b, ω_c and the temperature of the background radiation.

Mathematical theorems used in the treatise.

- M. Kac (1947): mean return time in a closed space.
- J. Bertrand (1873): only two force laws close every bound orbit.
- V. Fock (1935): Kepler orbits as great circles on the sphere S^3.
- P. A. M. Dirac (1931): product of electric and magnetic charge.
- B. Riemann (1859): mirror equation of the ζ-function.
- The modular group PSL(2,$\mathbb{Z}$) and Dedekind's η-function: symmetries of the torus shape.
- M. Morse: critical points and Euler characteristic.
- The duality of the force laws of Newton and Hooke, $(a + 3)(A + 3) = 4$.
- L. H. Thomas, E. Fermi (1927): the shell as a held funnel, $R \propto Z^{-1/3}$.
- J. H. Lane, R. Emden: the form $n = 3$ of the held funnel, $\omega_3 = 2.018$.
- L. Onsager, H. Casimir: symmetric and antisymmetric part of transport.
- A. S. Eddington: the radiation limit of a star.
- B. Carter (1974): stars at the border of radiation and convection, $\alpha_G \approx \alpha^{12} \cdot (m_e/m_p)^4$.

The author's source. Tabula Smaragdina (conclusion).